Notes on Stochastic Volatility Modelling

Kenneth Zhang

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Preface

A Note to the reader

There is something slightly theatrical about volatility.

Unlike price, which has the decency to reveal itself directly, volatility prefers inference, implication, and occasional misdirection. Entire markets quote and trade an object that nobody can observe, yet everyone insists they understand.

These notes were written in an effort to understand that object more honestly.

They owe much to a period spent in the company of practitioners for whom stochastic volatility was neither purely academic abstraction nor merely a pricing input, but a living geometry connecting derivatives, risk, and market behaviour. Exposure to such thinking leaves permanent side effects, including an unreasonable affection for Itô expansions and a tendency to view implied volatility surfaces with more sympathy than suspicion.

The treatment here is intentionally mechanical and mathematical. Most formulas are derived rather than presented, partly from conviction and partly from distrust. Markets may tolerate hand-waving; notes should not.

The reader will therefore encounter a generous amount of algebra, stochastic calculus, and occasionally the author's inability to leave a derivation unfinished.

About the Author

Kenneth Zhang is a quantitative researcher specializing in stochastic volatility, derivatives, and systematic trading research. His work spans volatility modeling, market microstructure, and machine learning applications in quantitative finance. Previously, he worked as a Quantitative Researcher within Systematic Volatility, where he studied options markets, volatility surfaces, and systematic derivatives strategies under the mentorship and guidance of Lorenzo Bergomi and Charbel Saad.

Kenneth's research interests lie at the intersection of stochastic calculus, volatility modeling, statistical learning, and market microstructure. These notes reflect both academic study and practical experience, with an emphasis on mathematical rigor, full derivations, and developing intuition through first principles rather than black-box results.

1 Black-Scholes PDE

1.1 Deriving the Black-Scholes PDE

For any pricing PDE to be valid, we require that the terminal condition be fulfilled for all S (underlying prices). That is, $P(t,S) = f(S)$ for all S where $f(\cdot)$ denotes the payoff function and $P(t,S)$ denotes the pricing model at time t with corresponding stock price S. More concretely, we require that $P(t=T, S_T) = f(S_T)$ for terminal price S_T. Assuming that we are short the option, then any change incurred over some small time interval δt delivers a negative value of our profit and loss (PnL).

Since we are short the option, we earn interest on the option's premium. That is, we earn $rP(t,S)\delta t$. To hedge against moves in the underlying, we buy Δ shares of the underlying, of which we finance at rate r. Therefore buying + financing the underlying hedging position contributed $\Delta\delta S - r\Delta S\delta t$ over the small time interval δt. We also earn money on dividends on the underlying given by q so the dividend paying component pays $q\Delta S\delta t$.

Thus, our overall PnL can be written as,

$$
\begin{aligned}
PnL &= -[P(t+\delta t, S+\delta t) - P(t,S)] + rP(t,S)\delta t + \Delta\delta S - r\Delta S\delta t + q\Delta S\delta t \\
&= -[P(t+\delta t, S+\delta S) - P(t,S)] + rP(t,S)\delta t + \Delta(\delta S - rS\delta t + qS\delta t) \\
&= -[\cancel{P(t,S)} + \underbrace{P_t\delta t + P_S\delta S + \frac{1}{2}P_{SS}(\delta S)^2}_{\text{Taylor Expansion of } P(t+\delta t, S+\delta S)} - \cancel{P(t,S)}] + rP\delta t + \Delta(\delta S - rS\delta t + qS\delta t) \\
&= -P_t\delta t - P_S\delta S - \frac{1}{2}P_{SS}(\delta S)^2 + rP\delta t + \Delta\delta S - r\Delta S\delta t + q\Delta S\delta t \\
&= \cancel{(\Delta - P_S)\delta S} + (-P_t + rP - r\Delta S + q\Delta S)\delta t - \frac{1}{2}P_{SS}(\delta S)^2, \quad \text{choose } \Delta := P_S \\
&= (-P_t + rP - rSP_S + qSP_S)\delta t - \frac{1}{2}P_{SS}(\delta S)^2 \\
&= -(P_t - rP + (r-q)SP_S)\delta t + \frac{1}{2}S^2P_{SS}\left(\frac{\delta S}{S}\right)^2. \qquad \text{(Equation 1.1)}
\end{aligned}
$$

The realized variance $\left(\frac{\delta S}{S}\right)^2$ over δt is of order $O(\sqrt{\delta t})$ so $\delta S = O(\sqrt{\delta t})$ meaning that $(\delta S)^2 = O(\delta t)$. Naturally, $\delta t\delta S$ is of order $O(\delta t^{3/2})$ meaning $(\delta t)^2 = O(\delta t)^2$ (read Appendix for full derivation). We keep our PDE of first order, so we only keep terms $P_t\delta$, $P_S\delta S$ and $\frac{1}{2}P_{SS}(\delta S)^2$ which after simplifying yields Equation 1.1.

Suppose we let $A(t,S)$ be the deterministic theta component $(P_t - rP + (r-q)SP_S)$ and let $B(t,S)$ be our gamma component $\frac{1}{2}S^2P_{SS}$. Clearly, if $A, B \geq 0$ then $-A\delta t \leq 0$ and $-Bx \leq 0$ for $x = (\delta S/S)^2$ meaning $PnL \leq 0$. Clearly, our pricing function will consistently underprice any option. Conversely, if $A, B \leq 0$, then $PnL \geq 0$ for all x meaning that our pricing function will consistently overprice all options.

We aim to deduce a implied break-even variance from this equation.

For our pricing function to be deemed usable, and more importantly, fair, we require that the breakeven volatility satisfies $PnL = 0$ that is $-A\delta t - B\left(\frac{\delta S}{S}\right)^2 = 0$. It follows that such break-even volatility is,

$$
\begin{aligned}
0 = -A\delta t - B\left(\frac{\delta S}{S}\right)^2 &\iff -A\delta t = B\left(\frac{\delta S}{S}\right)^2 \\
&\iff \left(\frac{\delta S}{S}\right)^2 = -\frac{A}{B}\delta t \\
&\iff \frac{\delta S}{S} = \pm\sqrt{-\frac{A(t,S)}{B(t,S)}\delta t} \in \mathbb{R} \\
&\iff -\frac{A}{B} > 0.
\end{aligned}
$$

It is important to note that we are working over arbitrarily small time intervals δt, so for some realized variance over the time interval $\left(\frac{\delta S}{S}\right)^2$ we estimate $\hat{\sigma}$ as the volatility level over δ. This is an approximation of the realized variance realized over the same interval i.e., $\left(\frac{\delta S}{S}\right)^2 \approxeq \hat{\sigma}^2 \delta t$.

Clearly, the condition that our $PnL = 0$ is satisfied if and only if $\left(\frac{\delta S}{S}\right)^2 \approxeq \hat{\sigma}^2 \delta t$.

Substituting realized volatility with our realized volatility estimate,

$$\begin{aligned} PnL = 0 &= -A\delta t - B\left(\frac{\delta S}{S}\right)^2 \\ &= -A\delta t - B\hat{\sigma}^2 \delta t \\ &= -(A + B\hat{\sigma}^2)\delta t \\ A &= -B\hat{\sigma}^2. \end{aligned}$$

Substituting this in with the definitions of A and B we get,

$$P_t - rP + (r-q)SP_S = -\hat{\sigma}^2 \frac{1}{2} S^2 P_{SS}$$

$$P_t - rP + (r-q)SP_S + \frac{1}{2}\hat{\sigma}^2 S^2 P_{SS} = 0. \qquad \text{(Equation 1.2)}$$

Equation 1.2 above is the famed *Black-Scholes PDE*.

1.2 Realized vs. Model Implied Break-Even Variance

Recall that we had previously written $A = -B\hat{\sigma}^2$. Substituting this in for A we get,

$$\begin{aligned} PnL &= -\left(-\hat{\sigma}^2 B\delta t\right) - B\left(\frac{\delta S}{S}\right)^2 \\ &= \hat{\sigma}^2 B\delta t - B\left(\frac{\delta S}{S}\right)^2 \\ &= -B\left[\left(\frac{\delta S}{S}\right)^2 - \hat{\sigma}^2 \delta t\right] \\ &= -\frac{1}{2} S^2 P_{SS} \left\{ \underbrace{\left(\frac{\delta S}{S}\right)^2}_{\text{realized variance}} - \underbrace{\hat{\sigma}^2 \delta t}_{\text{model-implied variance}} \right\}. \end{aligned}$$

This is simply the difference between the realized variance and the model implied break-even variance multiplied by the negative-half dollar-weighted gamma of the option (higher-order delta sensitivity). Since we are short the option, and we usually assume that $P_{SS} > 0$ and $\frac{1}{2} S^2 P_{SS} > 0$ meaning we lose money if the realised variance is greater than our implied and gain if the model implied is greater than the realised.

In general, we require that under the risk-neutral measure, the expected discounted future realized variance must equal the expected discounted value of the option's gamma-weighted exposure to instantaneous variance.

$$\mathbb{E}^{\mathbb{Q}}\left[\int_0^T e^{-rt} S_t^2 \sigma_t^2 dt\right] = \mathbb{E}^{\mathbb{Q}}\left[\int_0^t e^{-rt} S_t^2 \frac{\partial^2 P}{\partial S^2} \hat{\sigma}^2 dt\right].$$

Here, the term $S_t^2 \frac{\partial^2 P}{\partial S^2}$ denotes the option's dollar gamma exposure determining how sensitive the hedge is to quadratic variation / volatility. Therefore, the market's true instantaneous variance process σ_t^2 and option's price-implied/local volatility $\hat{\sigma}$, under no-arbitrage pricing, requires that the gamma-weighted average variance experienced by the option must match the gamma-weighted implied variance used in pricing.

1.3 Discretized PnL Attribution

In the real world, it is almost impossible for us to carry out perfect absolute continuous delta-hedging. However, suppose that we hedge at time t_i so that we instead discretize our PnL equation to the discrete delta-hedging case.

That is, instead of integrating over the discounted future variance, we use a summation as a discrete approximation of the integral. Suppose that instead we also observe returns r_i such that squaring gives us r_i^2.

Under Black-Scholes assumptions, our returns r_i can be written as $\sigma_i z_i$ so that $r_i^2 = \sigma_i^2 z_i^2$ where z_i's are iid standard normal random variables. Therefore, our PnL can be written as,

$$\begin{aligned} PnL &= -\sum_i e^{-rt_i} \frac{1}{2} S_i^2 P_{SS}^{\hat{\sigma}}(t_i, S_i)(r_i^2 - \hat{\sigma}^2 \delta t) \\ &= -\frac{1}{2} S_0^2 P_{SS}^{\hat{\sigma}} \sum_i (\sigma_i^2 z_i^2 - \hat{\sigma}^2)\delta t. \end{aligned}$$

Note that we have wrtten $e^{-rt_i} S_i^2 P_{SS}^{\hat{\sigma}}(t_i, S_i)$ as $S_0^2 P_{SS}^{\hat{\sigma}}$ because we are discounting to time $t = 0$. To be more explicit, we have written $P_{SS}^{\hat{\sigma}}$ to outline the dependence of P_{SS} on $\hat{\sigma}$. Notice that $-\frac{1}{2} S_0^2 P_{SS}^{\hat{\sigma}}$ is clearly deterministic.

We wish to make sense of $\sum_i (\sigma_i^2 z_i^2 - \hat{\sigma}^2)$ in our PnL.

Let's first compute the expectation of the sum,

$$\begin{aligned} \mathbb{E}\left[\sum_i (\sigma_i^2 z_i^2 - \hat{\sigma}^2)\right] &= \sum_i \left(\mathbb{E}\left[\sigma_i^2 z_i^2\right] - \hat{\sigma}^2\right)\delta t \\ &= \sum_i \left(\mathbb{E}\left[\sigma_i^2\right]\mathbb{E}\left[z_i^2\right] - \hat{\sigma}^2\right)\delta t \\ &= \sum_i \left(\mathbb{E}\left[\sigma_i^2\right] \cdot 1 - \hat{\sigma}^2\right)\delta t \quad \text{since } \mathbb{E}\left[z_i\right] = 1, \\ &= \sum_i (\hat{\sigma}^2 - \hat{\sigma}^2)\delta t \\ &= 0. \end{aligned}$$

Since the expectation of the sum is zero, the variance is simply computed as,

$$\begin{aligned} \mathbb{E}\left[\left[\sum_i (\sigma_i^2 z_i^2 - \hat{\sigma}^2)\delta t\right]^2\right] &= \mathbb{E}\left[\sum_{i,j} (\sigma_i^2 z_i^2 - \hat{\sigma}^2)(\sigma_j^2 z_j^2 - \hat{\sigma}^2)\delta t^2\right] \\ &= \sum_{i,j} \mathbb{E}\left[(\sigma_i^2 z_i^2 - \hat{\sigma}^2)(\sigma_j^2 z_j^2 - \hat{\sigma}^2)\delta t^2\right]. \end{aligned}$$

We will decompose this double sum into $i = j$ (diagonal) components and $i \neq j$ (off-diagonal) components.

Starting with the diagonal components,

$$\begin{aligned} (\sigma_i^2 z_i^2 - \hat{\sigma}^2)(\sigma_i^2 z_i^2 - \hat{\sigma}^2) &= (\sigma_i^2 z_i^2 - \hat{\sigma}^2) \\ &= \sigma_i^4 z_i^4 - 2\hat{\sigma}^2 \sigma_i^2 z_i^2 + \hat{\sigma}^4 \\ &\approxeq \mathbb{E}\left[\sigma_i^4 z_i^4\right] - 2\hat{\sigma}\mathbb{E}\left[\sigma_i^2 z_i^2\right] + \hat{\sigma}^4 \\ &= \mathbb{E}\left[\sigma_i^4 z_i^4\right] - 2\hat{\sigma}^4 \hat{\sigma}^4 + \hat{\sigma}^4 \\ &= \mathbb{E}\left[\sigma_i^4 z_i^4\right] - 2\hat{\sigma}^4 + \hat{\sigma}^4 \\ &= \mathbb{E}\left[\sigma_i^4 z_i^4\right] - \hat{\sigma}^4. \end{aligned}$$

We wish to make sense of $\mathbb{E}\left[\sigma_i^4 z_i^4\right]$. From the fourth moment we have that,

$$\mathbb{E}[X^4] = 3\sigma^4.$$

It follows for a standard normal random variable, κ controls the excess kurtosis,

$$\kappa = \frac{\mathbb{E}[r_i^4]}{\sigma^4} - 3 \implies \mathbb{E}[r_i^4] = (3+\kappa)\hat{\sigma}^4.$$

Thus we have that,

$$\mathbb{E}\left[\sigma_i^4 z_i^4\right] - \hat{\sigma}^4 = (3+\kappa-1)\hat{\sigma}^4 - \hat{\sigma}^4 = (2+\kappa)\hat{\sigma}^4,$$

for diagonal terms where $i = j$. For off-diagonal terms where $i \neq j$,

$$\begin{aligned}
\mathbb{E}\left[(\sigma_i^2 z_i^2 - \hat{\sigma}^2)(\sigma_j^2 z_j^2 - \hat{\sigma}^2)\right] &= \mathbb{E}\left[\sigma_i^2\sigma_j^2 z_i^2 z_j^2 - \hat{\sigma}^2\sigma_i^2 z_j^2 - \hat{\sigma}^2\sigma_j^2 z_j^2 + \hat{\sigma}^4\right] \\
&= \mathbb{E}\left[\sigma_i^2\sigma_j^2 z_i^2 z_j^2\right] - \hat{\sigma}^2\mathbb{E}\left[\sigma_i^2 z_i^2\right] - \hat{\sigma}^2\mathbb{E}\left[\sigma_j^2 z_j^2\right] + \hat{\sigma}^4 \\
&= \mathbb{E}\left[\sigma_i^2\sigma_j^2\right] - \hat{\sigma}^2\mathbb{E}\left[\sigma_i^2 z_i^2\right] - \hat{\sigma}^2\mathbb{E}\left[\sigma_j^2 z_j^2\right] + \hat{\sigma}^4 \\
&= \mathbb{E}\left[\sigma_i^2\sigma_j^2\right] - \hat{\sigma}^2\hat{\sigma}^2 - \cancel{\hat{\sigma}^2\hat{\sigma}^2 + \hat{\sigma}^4} \\
&= \mathbb{E}\left[\sigma_i^2\sigma_j^2\right] - \hat{\sigma}^4.
\end{aligned}$$

The expected value of the sum term is decomposed into a diagonal component and an off-diagonal component.

$$\mathbb{E}\left[\left[\sum_i (\sigma_i^2 z_i^2 - \hat{\sigma}^2)^2 \delta t^2\right]\right] = \sum_i (2+\kappa)\hat{\sigma}^4\delta t^2 + \sum_{i\neq j}\left(\mathbb{E}\left[\sigma_i^2\sigma_j^2\right] - \hat{\sigma}^4\right)\delta t^2.$$

The second component can be expressed with respect to a normalized auto-correlation term, or a two-time autocovariance term of instantaneous variance. We'll let the normalized form of this be denoted as,

$$f_{ij} = \frac{\left(\mathbb{E}\left[\sigma_i^2\sigma_j^2\right] - \hat{\sigma}^4\right)}{\mathbb{E}\left[\sigma^4\right] - \hat{\sigma}^4}.$$

We make sense of this autocorrelation term in the Appendix.

Thus, we write the term now as,

$$\begin{aligned}
\mathbb{E}\left[\left[\sum_i (\sigma_i^2 z_i^2 - \hat{\sigma}^2)^2 \delta t^2\right]\right] &= \sum_i (2+\kappa)\hat{\sigma}^4\delta t^2 + \sum_{i\neq j}\left(\mathbb{E}\left[\sigma^4\right] - \hat{\sigma}^4\right)\cdot f_{ij}\cdot\delta t^2 \\
&= \sum_i (2+\kappa)\hat{\sigma}^4\delta t + \left(\mathbb{E}\left[\sigma^4\right] - \hat{\sigma}^4\right)\sum_{i\neq j} f_{ij}\delta t^2 \\
&= \sum_i (2+\kappa)\hat{\sigma}^4\delta t^2 + \Omega\hat{\sigma}^4\sum_{i\neq j} f_{ij}\delta t^2, \quad \text{for } \Omega = \frac{\mathbb{E}\left[\sigma^4\right] - \hat{\sigma}^4}{\hat{\sigma}^4}, \\
&= \hat{\sigma}^4\left\{\sum_i (2+\kappa)\delta t^2 + \Omega\sum_{i\neq j} f_{ij}\delta t^2\right\}.
\end{aligned}$$

Therefore, the variance of our PnL can be expressed as,

$$\begin{aligned}
\text{Var}(PnL) &= \text{Var}\left(-\frac{1}{2}S_0^2 P_{SS}^{\hat{\sigma}}(t_0, S_0)\right)^2 \mathbb{E}\left[\sum_i (\sigma_i^2 z_i^2 - \hat{\sigma}^2)\delta t^2\right] \\
&= \left[\frac{1}{2}S_0^2 P_{SS}^{\hat{\sigma}}(t_0, S_0)\right]^2 \cdot \hat{\sigma}^4\left\{\sum_i (2+\kappa)\delta t^2 + \Omega\sum_{i\neq j} f_{ij}\delta t^2\right\}.
\end{aligned}$$

Before we compute the standard deviation, I will highlight an important expression. Recall that we can express our option's vega approximately as (read Appendix for full derivation),

$$\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} = S^2 \frac{\partial^2 P_{\hat{\sigma}}}{\partial S^2} \hat{\sigma} T.$$

Then it follows that we can write the following term as,

$$S^2 \frac{\partial^2 P_{\hat{\sigma}}}{\partial S^2} = \frac{1}{\hat{\sigma} T} \frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} \implies \frac{1}{2} \frac{1}{\hat{\sigma} T} \frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} = \frac{1}{2} S^2 \frac{\partial^2 P_{\hat{\sigma}}}{\partial S^2}.$$

Therefore, the standard deviation of our PnL can be computed as,

$$\begin{aligned}
SD(PnL) &= \sqrt{\mathrm{Var}(PnL)} \\
&= \sqrt{\left\{ \frac{1}{2} S_0^2 P_{SS}^{\hat{\sigma}}(t_0, S_0) \right\}^2 \cdot \hat{\sigma}^4 \left\{ \sum_i (2 + \kappa_i) \delta t^2 + \Omega \sum_{i \neq j} f_{ij} \delta t^2 \right\}} \\
&= \left| \frac{S_0^2}{2} P_{SS}^{\hat{\sigma}}(t_0, S_0) \right| \cdot \sqrt{\hat{\sigma}^4 \left\{ \sum_i (2 + \kappa_i) \delta t^2 + \Omega \sum_{i \neq j} f_{ij} \delta t^2 \right\}} \\
&= \left| \frac{S_0^2}{2} P_{SS}^{\hat{\sigma}}(t_0, S_0) \right| \cdot \hat{\sigma}^2 \sqrt{\sum_i (2 + \kappa_i) \delta t^2}, \quad \text{assuming uncorrelated increments,} \\
&= \left| \frac{S_0^2}{2} P_{SS}^{\hat{\sigma}}(t_0, S_0) \right| \cdot \hat{\sigma}^2 \sqrt{2 \sum_i \delta t^2}, \quad \text{assuming Normality so, } \kappa_i = 0, \\
&= \frac{1}{2} \cdot \left| S_0^2 P_{SS}^{\hat{\sigma}}(t_0, S_0) \right| \hat{\sigma}^2 \cdot \sqrt{2 \sum_i \delta t^2}, \\
&= \frac{1}{2} \left| \frac{1}{\hat{\sigma} T} \frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} \right| \hat{\sigma}^2 \cdot \sqrt{2 \sum_i \delta t^2} \\
&= \frac{1}{2T} \left| \hat{\sigma} \frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} \right| \cdot \sqrt{2 \sum_i \delta t^2} \\
&= \frac{1}{2T} \left| \hat{\sigma} \frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} \right| \cdot \sqrt{2 \cdot \frac{T^2}{N}} \\
&= \left| \frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} \right| \cdot \frac{\hat{\sigma}}{\sqrt{2N}}.
\end{aligned}$$

Thus, under Black-Scholes assumptions, a standard deviation of our PnL is equivalent to the impact of a relative volatility perturbation of size $\frac{1}{\sqrt{2N}}$. This follows because, if we view it as the option price impact caused by a volatility perturbation of size $\delta\hat{\sigma} = \frac{\hat{\sigma}}{\sqrt{2N}}$, then $\frac{\delta\hat{\sigma}}{\hat{\sigma}} = \frac{1}{\sqrt{2N}}$. To see this more clearly, we often use a standard realized variance estimator to estimate the volatility since $\hat{\sigma}^2$ is usually not available to us.

We often use the estimator,

$$\bar{\sigma}^2 = \frac{1}{N} \sum_i \left(\frac{S_{i+1} - S_i}{S_i} \right)^2 \approx \frac{1}{N} \hat{\sigma}^2 \sum_i z_i^2.$$

In Black-Scholes, we assume that the return is approximately Gaussian, that is $r_i \approx \hat{\sigma} z_i$ where $z_i \sim \mathcal{N}(0, 1)$.

Computing the expectation we get,

$$\mathbb{E}[\bar{\sigma}^2] \approx \mathbb{E}\left[\frac{\hat{\sigma}^2}{N} \sum_i z_i^2 \right] = \frac{\hat{\sigma}^2}{N} \sum_i \mathbb{E}[z_i^2] = \frac{\hat{\sigma}^2}{N} \sum_i 1 = \hat{\sigma}^2.$$

It is clear that this estimator is centered around the true variance and is therefore unbiased. We will not make statements on whether this is the best unbiased estimator for realized variance.

Now, we may compute the variance,

$$\text{Var}(\bar{\sigma}^2) \approx \text{Var}\left(\frac{\bar{\sigma}^2}{N}\sum_i z_i^2\right) = \frac{\hat{\sigma}^2}{N^2}\,\text{Var}\left(\sum_i z_i^2\right).$$

To find the variance of the independent z_i's, we use the 4th and 2 moments of z_i. Recall that for a standard normal variable z_i, $\mathbb{E}[z_i^4] = 3$ and $\mathbb{E}[z_i^2] = 1$. So, trivially $\text{Var}(z_i^2) = \mathbb{E}[z_i^4] - \mathbb{E}[z_i^2]^2 = 3 - 1 = 2$.

Therefore the variance of our estimator is simply,

$$\text{Var}(\bar{\sigma}^2) \approx \frac{\hat{\sigma}^4}{N^2}(2N) = \frac{2\hat{\sigma}^4}{N}.$$

Taking the root of our variance formulation above, we get the standard deviation of our estimator,

$$SD(\bar{\sigma}^2) = \hat{\sigma}^2\sqrt{\frac{2}{N}} = \hat{\sigma}^2 \cdot \sqrt{\frac{2}{N}} = \hat{\sigma}^2 \cdot \frac{\sqrt{2}}{\sqrt{N}} = \frac{1}{2}\cdot 2\hat{\sigma}^2 \cdot \frac{\sqrt{2}}{\sqrt{N}} = \frac{1}{2}\sqrt{\frac{2}{N}} \cdot (2\hat{\sigma}^2) = \frac{1}{\sqrt{2N}} \cdot (2\bar{\sigma}^2).$$

However, recall that volatility is the square root of variance so for small errors, if we let $V = \sigma^2$ then $\sigma = \sqrt{V}$. Thus a small change in $\frac{\delta\sigma}{\sigma} \approx \frac{1}{2}\frac{\delta V}{V}$. Thus, a relative standard deviation of the volatility estimator is approximately half the relative standard deviation of the variance estimator,

$$\frac{SD(\bar{\sigma})}{\hat{\sigma}} \approx \frac{1}{2}\cdot\frac{SD(\hat{\sigma}^2)}{\mathbb{E}[\hat{\sigma}^2]} = \frac{1}{2}\cdot\frac{\frac{1}{\sqrt{2N}}\cdot(2\hat{\sigma}^2)}{\mathbb{E}[\bar{\sigma}^2]} = \frac{1}{2}\frac{2\bar{\sigma}^2}{\sqrt{2N}\mathbb{E}[\hat{\sigma}^2]} = \frac{1}{2}\cdot\frac{2}{\sqrt{2N}}\cdot\frac{\hat{\sigma}^2}{\mathbb{E}[\hat{\sigma}^2]} \approxeq \frac{1}{2}\cdot\frac{2}{\sqrt{2N}}\cdot 1 = \frac{1}{\sqrt{2N}}.$$

Since we estimated the historical volatility then $SD(\hat{\sigma}_{\text{hist}}) \approx \frac{\hat{\sigma}}{\sqrt{2N}}$ which means a standard deviation in our PnL is equivalent to our option's vega times a perturbation in the volatility,

$$SD(PnL) = \frac{\hat{\sigma}}{\sqrt{2N}}\cdot\left|\frac{\partial P}{\partial\hat{\sigma}}\right| = \frac{1}{\sqrt{2N}}\left|\hat{\sigma}\cdot\frac{\partial P}{\partial\hat{\sigma}}\right|,$$

as we derived previously.

1.4 Pricing an ATM Call Option

Suppose we have a call with $S = 1, K = 1$ and time to maturity $T = 1$ with $r = q = 0$ and $\hat{\sigma} = 20\% = 0.20$.

Recall that the price of a European Call option is,

$$P = N(d_1) - Ke^{-rt}N(d_2) = N(d_1) - 1\cdot e^{-rt}N(d_2).$$

Solving for d_1 and d_2 explicitly we get,

$$d_1 = \frac{\frac{1}{2}\hat{\sigma}^2 T}{\hat{\sigma}\sqrt{T}} = \frac{1}{\hat{\sigma}\sqrt{T}}\cdot\frac{1}{2}\hat{\sigma}\cdot T = \frac{1}{2}\hat{\sigma}\sqrt{T} = 0.10, \qquad d_2 = 0.10 - 0.20 = -0.10.$$

Then, under constant volatility assumptions the Black-Scholes European call option premium is,

$$P = N(0.10) - N(-0.10) = N(0.10) - (1 - N(0.10)) = 2N(0.10) - 1 = 2\cdot 0.54 - 1 \approxeq 0.0796.$$

Under Black-Scholes assumptions, the option's premium is approximately 7.97% of the spot price.

Before we price the same European call option under stochastic volatility assumptions, we will make a claim of which I will prove here. We claim that $P \approxeq \hat{\sigma}\frac{\partial P}{\partial\hat{\sigma}}$ under short-maturity assumptions.

Proof. We claim that,

$$P_{\hat{\sigma}} \approxeq \frac{1}{\sqrt{2\pi}} S\hat{\sigma}\sqrt{T}, \qquad \text{(Bergomi 2016.)}$$

Differentiating this with respect to $\hat{\sigma}$ we get,

$$\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} \approxeq \frac{\partial}{\partial \hat{\sigma}} \left(\frac{1}{\sqrt{2\pi}} S\hat{\sigma}\sqrt{T} \right) = \frac{1}{\sqrt{2\pi}} S\sqrt{T}.$$

Multiplying the identity above by $\hat{\sigma}$ we get,

$$\hat{\sigma}\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} \approxeq \hat{\sigma} \left(\frac{1}{\sqrt{2\pi}} S\sqrt{T} \right) = \frac{1}{\sqrt{2\pi}} S\hat{\sigma}\sqrt{T} \approxeq P_{\hat{\sigma}}.$$

Thus for an ATM option, the approximation holds because the ATM option price is linear in volatility. □

Thus if we plug in $S = 1$ and $\hat{\sigma} = 0.20$ with $T = 1$,

$$P_{\hat{\sigma}} \approx \frac{1}{\sqrt{2\pi}} \cdot 1 \cdot 0.20 \cdot 1 = \frac{0.20}{\sqrt{2\pi}} \approx 0.0798.$$

This is very close to the exact Black-Scholes price of 7.97%.

What does the standard deviation of our PnL have to say about this price? If we plug in our values,

$$SD(PnL) = \frac{1}{\sqrt{2N}} \left| \hat{\sigma}\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} \right| = \frac{1}{\sqrt{2 \cdot 250}} \cdot \left| \hat{\sigma}\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} \right| \approx 0.045P.$$

Given our option premium is 7.97% of the spot price, the dollar relative to the spot price is $0.045P \approxeq 0.045 \times 0.0797 = 0.00359$. Therefore, the delta-hedged standard deviation of PnL is 0.36% of the spot. Relative to the premium, $\frac{SD(PnL)}{P} = 0.045$, that is 4.5% of the spot premium or approximately 5%.

For an ATM option, we also are familiar with the approximation that $P \approx C\hat{\sigma}$ where $C = \frac{1}{\sqrt{2\pi}} S\sqrt{T}$ meaning that a change in the option price is approximated by a change in volatility. So, a 4.5% relative uncertainty in volatility produces approximately a 5% relative uncertainty in the option price.

The statistical uncertainty of realized volatility is therefore,

$$\frac{SD(\hat{\sigma}_{\text{realized}})}{\hat{\sigma}} \approx \frac{1}{\sqrt{2N}} \approx 4.5\%.$$

Because an ATM option price is nearly linear in volatility we approximate,

$$\frac{SD(PnL)}{P} \approx \frac{SD(\hat{\sigma}_{\text{realized}})}{\hat{\sigma}} \approx 4.5\%.$$

Suppose that we have a fair option price of P and the risk from the uncertainty in realized volatility is $SD(PnL) \approx 0.05P$. If a dealer wants to change one standard deviation of protection on the offer, the ask price is approximately,

$$P_{\text{ask}} = P + 0.05P = 1.05P.$$

Alternatively, if the dealer wants a one standard deviation of protection on the bid, the bid price is approximately,

$$P_{\text{bid}} = P - 0.05P = 0.95P.$$

Therefore the bid/ask spread is,

$$P_{\text{ask}} - P_{\text{bid}} = 1.05P - 0.95P = 0.10P,$$

therefore the relative bid/ask spread is 10%.

1.5 Non-Black-Scholes PnL

Recall our formulation of the standard deviation of the PnL,

$$SD(PnL) = \left|\hat{\sigma}\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}}\right| \cdot \frac{1}{2T}\sqrt{\sum_i (2+\kappa)\delta t^2 + \Omega \sum_{i\neq j} f_{ij}\delta t^2}.$$

The term involving the excess kurtosis of our Gaussian returns perturbed by a Gaussian shock z_i is random. So, even if daily variances were constant, the realized squared return remains random due to the z_i's with κ adjusting for fatter/thinner tails. The second term, however, is the true stochastic volatility term. Empirically, daily returns are not independent, that is, if today's variance is high, tomorrow's variance is likely to be high too. So, the daily gamma/theta PnLs are correlated across time. Recall,

$$\Omega = \frac{\mathbb{E}\left[\sigma^4\right] - \hat{\sigma}^4}{\hat{\sigma}^4}, \qquad f_{ij} = \frac{\mathbb{E}\left[\sigma_i^2\sigma_j^2\right] - \hat{\sigma}^4}{\mathbb{E}\left[\sigma^4\right] - \hat{\sigma}^4},$$

where Ω measures dispersion of daily variances and f_{ij} measures correlation between variance at time i and variance at time j. We approximate the entire sum under time-homogeneous assumptions i.e., $\text{Cov}(\sigma_i^2, \sigma_j^2)$ only depends on time distance, as an integral,

$$\sum_{i\neq j} f_{ij}\delta t^2 \approxeq \int_0^T \int_0^T f(|t-u|)dtdu.$$

Note that the diagonals where $i = j$ have measure zero, therefore,

$$\begin{aligned}
\int_0^T \int_0^T f(|t-u|)dtdu &= 2\int_0^T \int_0^t f(t-u)dudt, \quad \text{symmetric about } [0,T]\times[0,T], \\
&= 2\int_0^T \int_0^t f(\tau)(-d\tau), \quad \tau = t-u, d\tau = -du, \\
&= 2\int_0^T \int_0^t f(\tau)d\tau dt, \quad 0 \leq \tau \leq t \leq T \iff 0 \leq \tau \leq t \wedge \tau \leq t \leq T \\
&= 2\int_0^T \int_\tau^T f(\tau)dtd\tau \\
&= 2\int_0^T f(\tau)(T-\tau)d\tau.
\end{aligned}$$

Moreover, since we are hedging N times over δt then $\delta = \frac{T}{N}$ so the first κ term becomes,

$$\sum_i (2+\kappa)\delta t^2 = (2+\kappa)\sum_i \delta t^2 = (2+\kappa)N\delta t^2 = (2+\kappa)N\left(\frac{T}{N}\right)^2 = (2+\kappa)\frac{T^2}{N}.$$

Thus our standard deviation formula of the PnL becomes,

$$\begin{aligned}
SD(PnL) &= \left|\hat{\sigma}\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}}\right| \cdot \frac{1}{2T}\sqrt{\sum_i (2+\kappa)\delta t^2 + \Omega \sum_{i\neq j} f_{ij}\delta t^2} \\
&\approxeq \left|\hat{\sigma}\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}}\right| \cdot \frac{1}{2T}\sqrt{(2+\kappa)\frac{T^2}{N} + 2\Omega\int_0^T (T-\tau)f(\tau)d\tau} \\
&= \left|\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}}\right| \sqrt{\frac{(2+\kappa)T^2/N}{4T^2} + \frac{2\Omega}{4T^2}\int_0^T (T-\tau)f(\tau)d\tau} \\
&= \left|\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}}\right| \sqrt{\frac{2+\kappa}{4N} + \frac{\Omega}{2T^2}\int_0^T (T-\tau)f(\tau)d\tau}.
\end{aligned}$$

From the formulation above, we are clearly able to recover the Black-Scholes PnL standard deviation equation,

$$\Omega = 0, \kappa = 0 \implies SD(PnL) = \left|\hat{\sigma}\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}}\right| \cdot \sqrt{\frac{2}{4N}} = \frac{1}{\sqrt{2N}}\left|\hat{\sigma}\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}}\right|.$$

In the case we have perfectly correlated daily variances i.e., where $f(\tau) = 1$ then,

$$\int_0^T (T-\tau)f(\tau)d\tau = \int_0^T (T-\tau)d\tau = \left[T\tau - \frac{\tau^2}{2}\right]_0^T = T^2 - \frac{T^2}{2} = \frac{T^2}{2}.$$

Therefore the stockastic volatility term under the square root becomes,

$$\frac{\Omega}{2T^2}\int_0^T (T-\tau)f(\tau)d\tau = \frac{\Omega}{2T^2}\cdot\frac{T^2}{2} = \frac{\Omega}{4}.$$

This term clearly does not vanish as the number of times we hedge goes to infinity. Therefore, more frequent delta-hedging does not remove volatility risk as $N \to \infty$. If only the second term mattered then,

$$SD(PnL) \approx \left|\hat{\sigma}\frac{\partial P}{\partial \hat{\sigma}}\right| \cdot \frac{\sqrt{\Omega}}{2}.$$

Since we know that an change in the option premium is approximately equivalent to a change in the option price with respect to a change in the volatility, then this is equivalent to a volatility perturbation of size,

$$\delta P \approx \frac{\partial P}{\partial \hat{\sigma}}\delta\hat{\sigma} \implies \delta\hat{\sigma} = \hat{\sigma}\frac{\sqrt{\Omega}}{2}.$$

This contribution is equivalent to mispricing $\hat{\sigma}$ by $\hat{\sigma}\frac{\sqrt{\Omega}}{2}$.

Daily returns in the real world are often not enough to cleanly separate into a variance component σ_i and a random shock piece z_i. Thus, we need to estimate daily volatility using intraday data. The daily variance is approximately $\sigma_i^2 = \sum_{\text{5-min intervals in day i}} r_{i,m}^2 + r^2_{\text{close-to-open, i}}$, from which we can compute the autocorrelation of daily variances. Empirically, daily variances are fit with an exponential,

$$f(\tau) = \rho e^{-k\tau}.$$

Clearly, $\rho < 1$ meaning that even at very short lags, the measured variance autocorrelation is below one. Using this empirical assumption, we can derive the explicit real-world formulation of the standard deviation of the PnL after removing Black-Scholes assumptions. Using our integral approximation we have,

$$\begin{aligned}
\int_0^T (T-\tau)f(\tau)\,d\tau &= \rho\left[T\frac{1-e^{-kT}}{k} - \int_0^T \tau e^{-k\tau}\,d\tau\right] \\
&= \rho\left[T\frac{1-e^{-kT}}{k} - \left(-\frac{T}{k}e^{-kT} + \frac{1-e^{-kT}}{k^2}\right)\right] \\
&= \rho\left[\frac{T}{k}(1-e^{-kT}) + \frac{T}{k}e^{-kT} - \frac{1-e^{-kT}}{k^2}\right] \\
&= \rho\left[\frac{T}{k} - \frac{T}{k}e^{-kT} + \frac{T}{k}e^{-kT} - \frac{1-e^{-kT}}{k^2}\right] \\
&= \rho\left[\frac{T}{k} - \frac{1-e^{-kT}}{k^2}\right] \\
&= \rho\left[\frac{kT}{k^2} - \frac{1-e^{-kT}}{k^2}\right] \\
&= \rho\left[\frac{kT-(1-e^{-kT})}{k^2}\right] \\
&= \rho\left[\frac{kT-1+e^{-kT}}{k^2}\right].
\end{aligned}$$

Plugging this into our vega-normalied standard deviation formulation of our PnL we have,

$$\begin{aligned}\frac{SD(PnL)}{\left|\hat{\sigma}\frac{\partial P}{\partial \hat{\sigma}}\right|} &= \sqrt{\frac{2+\kappa}{4N} + \frac{\Omega}{2T^2}\rho\frac{kT-1+e^{-kT}}{k^2}} \\ &= \sqrt{\frac{2+\kappa}{4N} + \frac{\rho\Omega}{2}\frac{kT-1+e^{-kT}}{k^2T^2}} \\ &= \sqrt{\frac{2+\kappa}{4N} + \frac{\rho\Omega}{2}\frac{kT-1+e^{-kT}}{(kT)^2}}.\end{aligned}$$

We see that this quantity is the relative volatility displacement that would produce the same option price change as one standard deviation of final delta-hedged PnL specifically from the term $\frac{\rho\Omega}{2}\frac{kT-1+e^{-kT}}{(kT)^2}$.

Using the stochastic volatility model of the standard deviation of PnL we can compute the same ATM European call option price. We'll use the estimates $\Omega = 2, \kappa = 5, \rho = 0.78$ and $\frac{1}{k} = 45$ days.

For a one-year option, there are approximately $N = 250$ trading days meaning that $T = 250$ days. Then $kT = \frac{250}{45} \approx 5.56$ so the dispersion term under the square root becomes,

$$\frac{2+\kappa}{4N} = \frac{2+5}{(4\times 250)} = \frac{7}{1000} = 0.007.$$

The stochastic volatility of variances becomes,

$$\frac{\rho\Omega}{2}\frac{kT-1+e^{-kT}}{(kT)^2} = \frac{0.78\cdot 2}{2}\cdot\frac{5.56-1+e^{-5.56}}{(5.56)^2} \approx 0.78\cdot\frac{4.56}{30.9} \approx 0.115.$$

The total inside the square root becomes 0.122 so $\sqrt{0.122} \approx 0.35$. So under stochastic volatility, since for an ATM option the option price is linear in volatility, a standard deviation of the PnL is equivalently 35% of the option premium. Black-Scholes showed that under the same ATM European call option, the PnL risk is only about 4.5% of the premium. Stochastic volatility is almost an entire order higher.

In the same one-year ATM call example, we showed that under Black-Scholes, the option premium was 7.97% of the spot, which meant that a standard deviation in the PnL is 0.36% of the spot. Using the stochastic volatility quantity, we see that in the real case, the standard deviation of the PnL is actually around $0.35 \times 0.0797 \approx 0.0279$ or 2.79% of the spot. This is again, almost an entire order greater in magnitude than Black-Scholes.

2 Local Volatility

2.1 Closed-Form form for ATM Implied Volatilities

Under the risk-neutral measure $\mathbb{Q}$, we assume the follow Black-Scholes stock price dynamics,

$$\frac{dS_t}{S_t} = (r-q)dt + \sigma dW_t^{\mathbb{Q}}.$$

We let $f(S_t) = \log S_t$ where $f(\cdot) = \log$. Applying Ito's Lemma using $f'(S_t) = \frac{1}{S_t}$ and $f''(S_t) = -\frac{1}{S_t^2}$,

$$\begin{aligned}
df(S_t) &\approxeq f'(S_t)dS_t + \frac{1}{2}f''(S_t)d(S_t)^2 \\
&= f'(S_t)dS_t + \frac{1}{2}f''(S_t)d\mathbb{E}\left[S_t\right] \\
&= \frac{1}{S_t}dS_t + \frac{1}{2}\left(-\frac{1}{S_t^2}\right)d\mathbb{E}\left[S_t\right] \\
&= \frac{dS_t}{S_t} - \frac{1}{2}\frac{d\mathbb{E}\left[S_t\right]}{S_t^2}.
\end{aligned}$$

We make sense of the quadratic variation term,

$$\begin{aligned}
\frac{d(S_t)^2}{S_t^2} &= \frac{d\langle S_t\rangle}{S_t^2} \\
&= \left\{(r-q)S_t dt + \sigma S_t dW_t^{\mathbb{Q}}\right\}^2 \\
&= (r-q)^2 S_t^2 \underbrace{(dt)^2}_{=0} + 2(r-q)\sigma S_t^2 \underbrace{dtdW_t^{\mathbb{Q}}}_{=0} + \sigma^2 S_t^2 \underbrace{(dW_t^{\mathbb{Q}})^2}_{=dt} \\
&= \sigma^2 S_t^2 dt.
\end{aligned}$$

Substituting the simplification of the quadratic variation term back into $d\log S_t$ we get,

$$\begin{aligned}
d\log S_t &= (r-q)dt + \sigma dW_t^{\mathbb{Q}} - \frac{1}{2}\frac{\sigma^2 S_t^2 dt}{S_t^2} \\
&= (r-q)dt + \sigma dW_t^{\mathbb{Q}} - \frac{1}{2}\sigma^2 dt \\
&= \left(r - q - \frac{1}{2}\sigma^2\right)dt + \sigma dW_t^{\mathbb{Q}}.
\end{aligned}$$

Taking logs over the closed form,

$$\begin{aligned}
\log S_T &= \log S_0 + \left\{r - q - \frac{1}{2}\sigma^2\right\}T + \sigma W_T^{\mathbb{Q}} \\
S_T &= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma W_T^{\mathbb{Q}}\right\} \\
&= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right\},
\end{aligned}$$

since under Normality assumption $W_T^{\mathbb{Q}} = \sqrt{T}Z$ where $Z \sim \mathcal{N}(0,1)$. $W_T^{\mathbb{Q}}$ is a Brownian motion.

Under risk-neutral measure, the price of a European call option is the discounted expectation under the risk-neutral measure of the option's payoff. In this case, the payoff of a European call option is $(S_T - K)^+$ so we derive algebraically,

$$\begin{aligned}
C^{\text{BS}}(S_0, K, T, \sigma) &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[(S_T - K)^+\right] \\
&= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[(S_T - K)\mathbf{1}\left\{S_T > K\right\}\right] \\
&= e^{-rT}\left\{\mathbb{E}^{\mathbb{Q}}\left[S_T\mathbf{1}\left\{S_T > K\right\}\right] - K\mathbb{Q}(S_T > K)\right\}.
\end{aligned}$$

We first wish to make sense of $\mathbb{E}^{\mathbb{Q}}\left[S_T \mathbf{1}\left\{S_T > K\right\}\right]$. Substituting the explicit form of S_T we can derive an explicit form for the comparison $S_T > K$,

$$\begin{aligned}
S_T > K &\iff S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}z\right\} > K \\
&\iff \log S_0 + \left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\cdot Z > \log(K) \\
&\iff \log\left(\frac{K}{S_0}\right) - \left(r - q - \frac{1}{2}\sigma^2\right) < \sigma\sqrt{T}Z \\
&\iff \frac{\log\left(\frac{K}{S_0}\right) - \left(r - q - \frac{1}{2}\sigma^2\right)}{\sigma\sqrt{T}} < Z.
\end{aligned}$$

If we let $d_2 = \frac{\log\left(\frac{S_0}{K}\right) + \left(r - q - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$ then it follows that $S_T > K$ is equivalent to $Z > -d_2$ which means that $\mathbb{Q}(S_T > K) = \mathbb{Q}(Z > -d_2) = N(-d_2)$. Since we have an explicit form for $S_T > K$, we will explicitly derive the entire term. We will let $a = \sigma\sqrt{T}$ for the sake of algebraic convenience.

$$\begin{aligned}
&\mathbb{E}^{\mathbb{Q}}\left[S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right\}\mathbf{1}\left\{Z > -d_2\right\}\right] \\
&= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T\right\}\mathbb{E}\left[\exp\left\{\sigma\sqrt{T}Z\right\}\mathbf{1}\left\{Z > -d_2\right\}\right] \\
&= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T\right\}\int_{-d_2}^{\infty} e^{az}\frac{1}{\sqrt{2\pi}}e^{-z^2/2}dz \\
&= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T\right\}\int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}}\exp\left\{az - \frac{1}{2}z^2\right\}dz \\
&= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T\right\}\int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{1}{2}\left(z^2 - 2az\right)\right\}dz \\
&= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T\right\}\int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{1}{2}(z-a)^2 + \frac{1}{2}a^2\right\}dz \\
&= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T\right\}e^{a^2/2}\int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{1}{2}(z-a)^2\right\}dz \\
&= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T\right\}e^{a^2/2}\int_{-d_2-a}^{\infty} \phi(u)du, \quad u := z - a \implies z = -d_2 \wedge u = -d_2 - a, \\
&= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T\right\}e^{a^2/2}N(d_2 + a) \\
&= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T\right\}e^{a^2/2}N(d_2 + \sigma\sqrt{T}).
\end{aligned}$$

Since we have d_2 then we can let d_1 be,

$$d_1 = \frac{\log\left(\frac{S_0}{K}\right) + \left(r - q - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}} + \sigma\sqrt{T} = \frac{\log\left(\frac{S_0}{K}\right) + \left(r - q - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}} = d_2 + \sigma\sqrt{T},$$

which is precisely the term inside $N(d_2 + a)$. Therefore we have an explicit closed-form solution,

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}\left[S_T \mathbf{1}\left\{S_T > K\right\}\right] &= S_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T\right\}\exp\left\{\frac{1}{2}\sigma^2 T\right\}N(d_1) \\
&= S_0 \exp\left\{(r - q)\,T\right\}N(d_1).
\end{aligned}$$

Trivially, it follows that the Black-Scholes price for a European call option is,

$$\begin{aligned}
C^{\mathrm{BS}}(S_0, K, T, \sigma) &= e^{-rT}\left[S_0 \exp\left\{(r-q)T\right\}N(d_1) - KN(d_2)\right] \\
&= e^{-rT}e^{(r-q)T}S_0 N(d_1) - Ke^{-rT}N(d_2) \\
&= S_0 e^{-qT}N(d_1) - Ke^{-rT}N(d_2).
\end{aligned}$$

We wish to calibrate the implied surface of the Black-Scholes European call option price to the market call option prices, or in other words, solve the equality for a $\sigma_{\text{imp}}(K,T)$ such that,

$$C^{\text{BS}}(S_0, K, T, \sigma_{\text{imp}}(K,T)) = C^{\text{mkt}}(K,T).$$

We will define the notation $c(0,T,x,K)$. Suppose $x = S_0$, then we can write the market call option surface as $c(0,T,x,K) = e^{-rT}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ \mid x = S_0]$ representing the current market call option price with maturity T, at stock stock price x and strike K as the discounted risk-neutral expected payoff.

Consider a European Call option with payoff $(S_T - K)^+$ which is in-the-money if and only if $S_T > K$. Under log-dynamics, recall that the terminal price of the stock price S_t is,

$$S_T = S_0 \exp\left[\left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma W_T^{\mathbb{Q}}\right] = S_0 \exp\left[\left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right].$$

Notice that we have re-written the Brownian-motion under risk-neutral probability measure $\mathbb{Q}$ as $\sqrt{T}$ as a Brownian motion has normally distributed increments with mean 0 and constant variance. We wish to quantify the probability for when $S_T > K$ or when the European Call option is in-the-money. Using the definition of the terminal stock/forward price, we can derive a closed form formulation for $S_T > K$.

We start with the terminal condition $S_T > K$. We will use forward price notation to be more consistent with literature. For a forward contract with payoff $(F_T - K)^+$, the forward contract is in-the-money if and only if $F_T > K$ holds at terminal time $t = T$. Therefore we can write,

$$\begin{aligned}
S_T > K &\iff F_T > K \\
&\iff F_0 \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right\} > K \\
&\iff \exp\left\{\left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right\} > \frac{K}{F_0}.
\end{aligned}$$

We can re-write the Black-Scholes equation so that it models the current futures/forward price, in a formulation commonly referred to as the Black-76 model where $F_t(T)$ is the futures or forward price at time t for delivery at date T. Suppose that we let $\tau = T - t$ be the time to maturity until the option expires. For a European call option on a futures contract, $C_t = e^{-r\tau}\left[F_t N(d_1) - K N(d_2)\right]$. Here F_t is the current futures prices and K is the option strike with implied volatility σ of the futures price and r being the continuously compounded interest rate used to discount the option payoff. Notice that r does not appear inside d_1 and d_2 so it appears as the Black-Scholes case when $r = q = 0$. However, it is important to note that the interest rate r is still present in the discount factor $e^{-r\tau}$. In contracts such as SPXW 0DTE options where time to maturity is near zero (since time to maturity is often measured in minutes or seconds given the short-datedness of the option), the discounting term can be option to include as it's value is negligible.

For reader's who are not familiar with financial notation, the spot price is necessarily differentiated from the forward price. Suppose that a stock ius worth $S_t = 100$ and the risk-free rate is 5% with dividend yield 2% and maturity is one year away. The forward price is $F_t(T) = 100e^{(0.05-0.02)} \approx 103.05$. In the spot Black-Scholes formulation, the option model starts from $S_t = 100$ and explicitly includes $r - q = 3\%$ in the distribution. In Black-76, the model starts directly from $F_t(T) = 103.05$, eliminating the need to add another $r - q$ drift term. Doing so would count the cost of the carry twice, but we still need to discount the option's maturity payoff.

Continuing, from Black-76 we have that,

$$\begin{aligned}
\left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z &> \log\left(\frac{K}{F_0}\right) \\
-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}Z &> \log\left(\frac{K}{F_0}\right).
\end{aligned}$$

Therefore the condition simplifies to,

$$\begin{aligned}\sigma\sqrt{T}Z &> \log\left(\frac{K}{F_0}\right) + \frac{1}{2}\sigma^2 T \\ Z &> \frac{\log\left(\frac{K}{F_0}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \\ Z &> \frac{\log\left(\frac{K}{S_0}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}.\end{aligned}$$

By definition of d_2, we have translated $S_T > K$ into the probability $Z > -d_2$ so $\mathbb{P}(S_T > K)$ under Black-Scholes dynamics is the same as writing $\mathbb{P}(Z > -d_2)$ under risk-neutral probabilities.

Recalling from earlier that from the identity above, we have the price of a European Call option under Black-Scholes dynamics so we can write,

$$\begin{aligned}C_{\text{BS}} &= e^{-rT}\left\{\mathbb{E}[S_T \mathbf{1}\left\{S_T > K\right\}] - K\mathbb{P}(Z > -d_2)\right\} \\ &= e^{-rT}\left\{S_0 e^{(r-q)T}N(d_1) - K\mathbb{P}(Z > -d_2)\right\} \\ &= S_0 e^{-qT} - Ke^{-rT}N(d_2).\end{aligned}$$

Clearly the term involving the strike K of the European Call option simplifies because,

$$K\mathbb{P}(Z > -d_2) = K\left[1 - N(-d_2)\right] = KN(d_2).$$

It is convenient to derive the Put-Call parity under Black-76 as well as,

$$\begin{aligned}C - P &= e^{-rT}[S_T - K] \\ P &= C - e^{r-T}[S_T - K] \\ &= e^{-rT}\left\{S_0 e^{-qT}N(d_1) - KN(d_2)\right\} - e^{-rT}[S_T - K] \\ &= e^{-rT}\left\{S_0 N(d_1) - KN(d_2) - S_T + K\right\} \\ &= e^{-rT}\left\{K(1 - N(d_2)) - S_T(1 - N(d_2))\right\} \\ &= e^{-rT}\left\{KN(-d_2) - S_T N(-d_1)\right\} \\ &= e^{-rT}\left[KN(-d_2) - FN(-d_1)\right].\end{aligned}$$

Let's take a step back for a second. Recall that the implied volatility $\sigma_{\text{imp}}(K,T)$ maps Black-Scholes call option prices to market call option prices for a given strike and maturity. Therefore, the true formulation satisfies $C_{\text{BS}}(S_0, K, T, \sigma_{\text{imp}}(K,T)) = C_{\text{mkt}}(K,T)$. Let $D = e^{-rt}$ be our discount factor so that we can write,

$$\frac{C_{\text{mkt}}(K,T)}{D} = \frac{D(FN(d_1) - K\phi(d_2))}{D} = FN(d_1) - KN(d_2).$$

In order to obtain the implied volatility, we often require numerical inversion methods that do not have a closed-form solution. Suppose that we define $v = \sigma_{\text{imp}}\sqrt{T}$ and log-moneyness $x = \log\left(\frac{F}{K}\right)$. Then,

$$d_1 = \frac{x}{v} + \frac{v}{2}, \qquad d_2 = \frac{x}{v} - \frac{v}{2},$$

so that the call market price relative to the discount rate can be written as,

$$\frac{C_{\text{mkt}}}{D} = FN\left(\frac{x}{v} + \frac{v}{2}\right) - KN\left(\frac{x}{v} - \frac{v}{2}\right).$$

A typical Newton's method scheme for obtaining the implied volatility could be,

$$\begin{aligned}\sigma_{n+1} &= \sigma_n - \frac{C_{\text{BS}}(\sigma) - C_{\text{mkt}}}{\frac{\partial C_{\text{BS}}}{2\sigma}(\sigma_n)} \\ &= \sigma_n - \frac{D(F\phi(d_1) - K\phi(d_2)) - C_{\text{mkt}}}{DF\phi'(d_1)\sqrt{T}}.\end{aligned}$$

Note that in this case, $\frac{\partial C_{\text{BS}}}{2\sigma}$ is simply the option's vega which can be sufficiently written as $DF\phi'(d_1)\sqrt{T}$. While a closed-form cannot be obtain for the general case, we certainly can obtain a closed-form for when the option is at-the-money i.e., when $K \approxeq F$. When $F \approxeq K$, we know that $\frac{F}{K} \approxeq 1$ which means that the moneyness $\log\left(\frac{F}{K}\right) \approx 0$ so $\frac{x}{v} \approx 0$ where $x = \log\left(\frac{F}{K}\right)$. Then, the market call price with respect to discounting is,

$$\begin{aligned}\frac{C_{\text{mkt}}}{D} &= F\phi\left(\frac{v}{2}\right) - F\phi\left(-\frac{v}{2}\right) \\ &= F\left\{\phi\left(\frac{v}{2}\right) - \left(1 - \phi\left(\frac{v}{2}\right)\right)\right\}.\end{aligned}$$

Dividing by the forward price we see that,

$$\frac{C_{\text{mkt}}}{DF} = 2\phi\left(\frac{v}{2}\right) - 1.$$

Solving for $v = \sigma_{\text{imp}}\sqrt{T}$ we have that,

$$\begin{aligned}\phi\left(\frac{v}{2}\right) &= \frac{C_{\text{mkt}}}{DF} \cdot \frac{1}{2} + 1 \\ \frac{v}{2} &= \phi^{-1}\left\{\frac{1}{2}\left(1 + \frac{C_{\text{mkt}}}{DF}\right)\right\} \\ v &= 2\phi^{-1}\left\{\frac{1}{2}\left(1 + \frac{C_{\text{mkt}}}{DF}\right)\right\}.\end{aligned}$$

Adjusting for maturity we see that this is simply,

$$\sigma_{\text{imp}} = \frac{2}{\sqrt{T}} \cdot \phi^{-1}\left\{\frac{1}{2}\left(1 + \frac{C_{\text{mkt}}}{DF}\right)\right\}.$$

Using forward linear approximations we see that $2\phi\left(\frac{v}{2}\right) - 1 \approxeq \frac{v}{\sqrt{2\pi}}$ which means that,

$$\frac{C_{\text{mkt}}}{DF} \approxeq \frac{\sigma_{\text{imp}}\sqrt{T}}{\sqrt{2\pi}} \implies \sigma_{\text{imp}} \approxeq \sqrt{\frac{2\pi}{T}}\frac{C_{\text{imp}}}{DF} \approxeq \sqrt{\frac{2\pi}{T}} \cdot \frac{P_{\text{mkt}} + D(F-K)}{DF},$$

by Put-Call parity and first-order linear approximation.

2.2 Dupire's Local Volatility Model

Consider a European Call option with payoff $f(S_T) = (S_T - K)^+$. Suppose that over a small interval $[y, y + \delta y]$ on $[0, T]$ we observe the underlying stock price change S_t to $S_t + \frac{dS_t}{S_t}$. The probability density that the stock price is at some stock price S_T over a sufficiently small interval $dy = \delta y$ given initial stock price $S_0 = x$ is given by the transition density,

$$\mathbb{P}(S_T \in dy \mid S_0 = x) = \tilde{p}(0, T, x, y)dy \approxeq \mathbb{P}(S_T \in [y, y + dy] \mid S_0 = x).$$

where the right-hand side of the equality above is simply the transition density of the stock price being at value y at some time T given that it was at x at initial time $t = 0$ over some small interval dy. More simply put, we start at x at time $t = 0$ and end up at y at time $t = T$, according to the transition density $\tilde{p}$. We write this density under the risk-neutral probability measure $\mathbb{Q}$.

Given this notation, we can therefore the write the transition state of the European call price under risk-neutral measure $\mathbb{Q}$ as the discounted expectation of the call option payoff conditional on S_0 being at level x.

$$\begin{aligned} C(0,T,x,K) &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[(S_T-K)^+ \mid S_0 = x\right] \\ &= e^{-rT}\int_0^{\infty}(y-K)^+\tilde{p}(0,T,x,y)dy \\ &= e^{-rT}\int_K^{\infty}(y-K)\tilde{p}(0,T,x,y)dy. \end{aligned}$$

The last line sets the lower bound to K since that is the lower bound of the at-the-money to in-the-money region. We do not consider when the European call option is out-the-money since it eliminates the transition density.

Similarly, over a small interval dy we can write the European put call payoff as,

$$\begin{aligned} P(0,T,x,K) &= e^{-rT}\int_0^{\infty}(K-y)^+\tilde{p}(0,T,x,y)dy \\ &= e^{-rT}\int_0^{K}(K-y)\tilde{p}(0,T,x,y)dy. \end{aligned}$$

Over a small horizon h, using Chapman-Kolmogorov, we can write the transition density as,

$$p(T,y)dy = \mathbb{P}^{\mathbb{Q}}\left(S_T \in [y,y+dy] \mid S_0 = x\right), \qquad p(T+h,y)dy = \int_0^{\infty} p(T,z)p(T,z;T+h,y)dz.$$

Therefore the expectations of our payoffs becomes,

$$\mathbb{E}[\phi(S_T) \mid S_0 = x] = \int_0^{\infty}\phi(y)p(T,y)dy, \qquad \mathbb{E}[\phi(S_{T+h}) \mid S_0 = x] = \int_0^{\infty}\phi(y)p(T+h,y)dy.$$

We wish to make sense of what these expectations mean. Applying Chapman-Kolmogorov again, we obtain,

$$\begin{aligned} \mathbb{E}[\phi(S_{T+H}) \mid S_0 = x] &= \int_0^{\infty}\phi(y)p(T+h,y)dy \\ &= \int_0^{\infty}\phi(y)\left[\int_0^{\infty}p(T,z)p(T,z;T+h,y)dy\right] \\ &= \int_0^{\infty}p(T,z)\left[\int_0^{\infty}\phi(y)p(T,z;T+h,y)dy\right]dz \\ &= \int_0^{\infty}p(T,z)\mathbb{E}^{\mathbb{Q}}\left[\phi(S_{T+h}) \mid S_T = x\right]dz. \end{aligned}$$

The final line represents the expected payoff at $T+h$ is the average over all possible values z of S_T of the conditional expected payoff starting from $S_T = z$ i.e., an application of the Markov property and Law of Iterated Expectations. Substituting our payoff function for $\phi(S_T)$ we get that,

$$\begin{aligned} \mathbb{E}[(S_T-K)^+] &= \int_{-\infty}^{\infty}(S_T-K)^+\cdot\tilde{p}(0,T,x,S_T)dS_T \\ &= \int_K^{\infty}(y-K)^+\tilde{p}(0,T,x,y)dy. \end{aligned}$$

The discounted expected payoff becomes,

$$D\cdot\mathbb{E}[(S_T-K)^+] = e^{-rT}\mathbb{E}[(S_T-K)^+] = e^{-rT}\int_K^{\infty}(y-K)^+\tilde{p}(0,T,x,y)dy.$$

To simplify this risk-neutral discounted expectation, we first need to derive some asymptotic equations of the underlying stock dynamics. Suppose that we let $\sigma_{\text{loc}}(t,S_t)$ be the local variance with respect to a sample path S_t of the underlying stock price. Then our stock price SDE becomes,

$$dS_t = (r-q)S_t dt + \sigma_{\text{loc}}(t,S_t)S_t dW_t^{\mathbb{Q}}.$$

Over all a sufficiently small horizon h, the change in the stock price over the horizon can simply be written as,

$$S_{T+h} = S_T + (r-q)S_T \cdot h + \sigma_{\text{loc}}(T, S_T)S_T \cdot \left(W^{\mathbb{Q}}_{T+h} - W^{\mathbb{Q}}_T\right).$$

By properties of Brownian motion we can write,

$$\Delta S = (r-q)S_T \cdot h + \sigma_{\text{loc}}(T, S_T)S_T \Delta W.$$

Conditioning on $S_T = x$, the risk-neutral expectation of the change in the stock price over h becomes,

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}[\Delta S \mid S_T = x] &= \mathbb{E}^{\mathbb{Q}}\left[(r-q)zh + \sigma_{\text{loc}}(T,z)z \cdot \Delta W \mid S_T = z\right] \\
&= (r-q)zh + \sigma_{\text{loc}}(T,z)z \cdot \underbrace{\mathbb{E}[\Delta W]}_{=0} \\
&= (r-q)zh.
\end{aligned}$$

The second moment of ΔS therefore follows as,

$$\begin{aligned}
&\mathbb{E}^{\mathbb{Q}}[(\Delta S)^2 \mid S_T = x] \\
&= \mathbb{E}\left[(r-q)^2 z^2 h^2 + 2(r-q)zh \cdot \sigma_{\text{loc}}(T,z)z\Delta W + \sigma^2_{\text{loc}} z^2 (\Delta W)^2\right] \\
&= (r-q)^2 z^2 h^2 + 2(r-q)\sigma_{\text{loc}}(T,z)z^2 h\mathbb{E}[\Delta W] + \sigma^2_{\text{loc}}(T,z)z^2\mathbb{E}\left[(\Delta W)^2\right] \\
&= (r-q)^2 z^2 h^2 + \sigma^2_{\text{loc}}(T,z)z^2 h.
\end{aligned}$$

We use a Taylor expansion up to the second order for the discounted risk-neutral expectation of our payoff,

$$\begin{aligned}
&\mathbb{E}^{\mathbb{Q}}[\phi(S_{T+h}) \mid S_T = z] \\
&= \mathbb{E}^{\mathbb{Q}}[\phi(z + \Delta S) \mid S_T = z] \\
&= \mathbb{E}^{\mathbb{Q}}\left[\phi(z) + \phi'(z)\Delta S + \frac{1}{2}\phi''(z)(\Delta S)^2 \mid S_T = Z\right] \\
&= \phi(z) + \phi'(z)\mathbb{E}[\Delta S \mid S_T = z] + \frac{1}{2}\phi''(z)\mathbb{E}^{\mathbb{Q}}[(\Delta S)^2 \mid S_t = z] \\
&= \phi(z) + \phi'(z)\mathbb{E}[(r-q)zh + o(h)] + \frac{1}{2}\phi''(z)\left[\sigma^2_{\text{loc}}(T,z)z^2 h + o(h)\right] \\
&= \phi(z) + h(r-q)z\phi'(z) + \frac{1}{2}h\sigma^2_{\text{loc}}(T,z)z^2\phi''(z) + o(h).
\end{aligned}$$

So conditioning on some intermediary stock price,

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}\left[\phi(S_{T+h}) \mid S_0 = x\right] &= \int_0^\infty \phi(y)p(T+h, y)dy \\
&= \int_0^\infty \phi(y)\left\{\int_0^\infty p(T,z)p(T,z;T+h,y)dz\right\}dy \\
&= \int_0^\infty p(T,z)\left[\int_0^\infty \phi(y)p(T,z;T+h,y)dy\right]dz \\
&= \int_0^\infty p(T,z)\mathbb{E}^{\mathbb{Q}}\left[\phi(S_{T+h}) \mid S_T = z\right]dz.
\end{aligned}$$

Note that the last line follows by *Law of Iterated Expectations* where together with the Markov property, we can break the evolution from 0 to $T+h$ into two stages, that is $0 \to T$, $T \to T+h$.

More precisely we have that,

$$\mathbb{E}^{\mathbb{Q}}[\phi(S_{T+h}) \mid S_0 = x] = \mathbb{E}^{\mathbb{Q}}\left[\mathbb{E}^{\mathbb{Q}}[\phi(S_{T+h}) \mid S_T]\middle| S_0 = x\right].$$

Note here that z is a possible value of S_T and $p(0,x;T,z)dz$ is the probability that S_T lies near z given that $S_0 = x$. Naturally, $\mathbb{E}^{\mathbb{Q}}[\phi(S_{T+h}) \mid S_T = z]$ is the expected payoff over the remaining short interval $[T, T+h]$ assuming the process is at z at time T. This can be interpreted as calculating the final expected payoff via averaging the short-horizon value from T to $T+h$ over all possible values of S_T.

Substituting our second-order Taylor expansion we get,

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}\left[\phi(S_{T+h}) \mid S_0 = x\right] &= \int_0^\infty \left[\phi(z) + h(r-q)z\phi'(z) + \frac{1}{2}h\sigma_{\text{loc}}^2(T,z)z^2\phi'(z) + o(h)\right] p(T,z)dz \\ &= \int_0^\infty p(T,z)dz + h\int_0^\infty (r-q)z\phi'(z)p(T,z)dz \\ &\quad + \frac{1}{2}h\int_0^\infty \sigma_{\text{loc}}^2(T,z)z^2\phi''(z)p(T,z)dz + o(h).\end{aligned}$$

Note that if we take differences between time T and $T+h$, the difference is simply,

$$\begin{aligned}D(T,T+h) &= \int_0^\infty \phi(y)p(T+h,y)dy - \int_0^\infty \phi(y)p(T,y)dy \\ &= h\int_0^\infty (r-q)z\phi'(z)p(T,z)dz + \frac{1}{2}h\int_0^\infty \sigma_{\text{loc}}^2(T,y)y^2\phi''(y)p(T,y)dy.\end{aligned}$$

Letting $h \to 0$ we see that,

$$\begin{aligned}&\lim_{h\to 0}\frac{1}{h}\left[\int_0^\infty \phi(y)p(T+h,y)dy - \int_0^\infty \phi(y)p(T,y)dy\right] \\ &\quad= \lim_{h\to 0}\left[\int_0^\infty (r-q)z\phi'(z)p(T,z)dz + \frac{1}{2}\int_0^\infty \sigma_{\text{loc}}^2(T,y)y^2\phi''(y)p(T,y)dy\right] \\ &\quad= \frac{\partial}{\partial T}\int_0^\infty \phi(y)p(T,y)dy \\ &\quad= \int_0^\infty \phi(y)p_T(T,y)dy.\end{aligned}$$

Simplifying and integrating by parts gives us,

$$\begin{aligned}&= \int_0^\infty \phi'(y)\left[(r-q)yp(T,y)\right]dy + \frac{1}{2}\int_0^\infty \phi''(y)\sigma_{\text{loc}}^2(T,y)y^2p(T,y)dy \\ &= \left[\phi(y)(r-q)yp(T,y)\right]_0^\infty - \int_0^\infty \phi'(y)\frac{\partial}{\partial y}\left[(r-q)yp(T,y)\right]dy \\ &\quad \frac{1}{2}\left\{\left[\phi'(y)\sigma_{\text{loc}}^2(T,y)y^2p(T,y)\right]_0^\infty - \int_0^\infty \phi'(y)\frac{\partial}{\partial y}\sigma_{\text{loc}}^2(T,y)y^2p(T,y)dy\right\} \\ &= \cancel{\left[\phi(y)(r-q)yp(T,y)\right]_0^\infty} - \int_0^\infty \phi(y)\frac{\partial}{\partial y}\left[(r-q)yp(T,y)\right]dy \\ &\quad \frac{1}{2}\left\{\cancel{\left[\phi'(y)\sigma_{\text{loc}}^2(T,y)y^2p(T,y)\right]_0^\infty} - \int_0^\infty \phi'(y)\frac{\partial}{\partial y}\sigma_{\text{loc}}^2(T,y)y^2p(T,y)dy\right\} \\ &= -\int_0^\infty \phi(y)\frac{\partial}{\partial y}(r-q)yp(T,y)dy - \frac{1}{2}\int_0^\infty \phi'(y)\frac{\partial}{\partial y}\sigma_{\text{loc}}^2(T,y)y^2p(T,y)dy \\ &= -\int_0^\infty \phi(y)\frac{\partial}{\partial y}(r-q)yp(T,y)dy \\ &\quad -\frac{1}{2}\left\{\cancel{\left[\phi(y)\frac{\partial}{\partial y}\sigma_{\text{loc}}^2(T,y)y^2p(T,y)\right]_0^\infty} - \int_0^\infty \phi(y)\frac{\partial^2}{\partial y^2}\sigma_{\text{loc}}^2(T,y)y^2p(T,y)dy\right\} \\ &= -\int_0^\infty \phi(y)\frac{\partial}{\partial y}(r-q)yp(T,y)dy + \frac{1}{2}\int_0^\infty \phi(y)\left[\frac{\partial^2}{\partial y^2}\sigma_{\text{loc}}^2(T,y)y^2p(T,y)\right]dy \\ &= \int_0^\infty \phi(y)\left[-\frac{\partial}{\partial y}\left[(r-q)yp(T,y)\right] + \frac{1}{2}\frac{\partial^2}{\partial y^2}\left[\sigma_{\text{loc}}^2(T,y)y^2p(T,y)\right]\right]dy \\ &= \int_0^\infty \phi(y)\left[p_T(T,y) + \frac{\partial}{\partial y}\left[(r-q)yp(T,y)\right] - \frac{\partial^2}{\partial y^2}\left[\sigma_{\text{loc}}^2(T,y)y^2p(T,y)\right]\right]dy.\end{aligned}$$

Note that for a sufficiently smooth payoff function ϕ the expression inside the brackets must be zero so,

$$\int_0^\infty \phi(y)\left[p_T(T,y) + \frac{\partial}{\partial y}\left[(r-q)yp(T,y)\right] - \frac{\partial^2}{\partial y^2}\left[\sigma_{\text{loc}}^2(T,y)y^2p(T,y)\right]\right]dy = 0.$$

Therefore we conclude that the time derivative of our transition density at (T, y) is simply,

$$p_T(T,y) = -\frac{\partial}{\partial y}\left[(r-q)yp(T,y)\right] + \frac{1}{2}\frac{\partial^2}{\partial y^2}\left[\sigma_{\text{loc}}^2(T,y)y^2p(T,y)\right].$$

Recall that Dupire's central claim is that,

> *Given a sufficiently smooth, arbitrage-free surface of European call prices across all strikes K and maturities T, one can recover a local volatility function $\sigma_{loc}(T,S)$ that exactly reproduces that entire option-price surface. That is, the slope, curvature, and maturity evolution of the call-price surface determine local volatility.*

Recall that a European call option price can be written as,

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}\left[(S_T-K)^+ \mid S_0 = x\right] &= \int_0^\infty (y-K)^+ \cdot \tilde{p}(0,T,x,y)dy \\
&= \int_0^K (y-K)\tilde{p}(0,T,x,y)dy + \int_K^\infty (y-K)^+ \cdot \tilde{p}(0,T,x,y)dy \\
&= \int_K^\infty (y-K)^+\tilde{p}(0,T,x,y)dy,
\end{aligned}$$

since the first integral on the second line is zero since the European call option is out-the-money. To determine the local volatility function according to Dupire's thesis, we determine C_K, C_{KK}, and C_T from $C(0,T,x,K)$,

$$C(0,T,x,K) = e^{-rT}\int_0^\infty (y-K)^+ \cdot \tilde{p}(0,T,x,y)dy.$$

The derivative with respect to strike K is quite trivial,

$$\begin{aligned}
C_K &= e^{-rT}\frac{\partial}{\partial K}\int_K^\infty (y-K)\tilde{p}(0,T,x,y)dy \\
&= e^{-rT}\left\{\left[-(y-K)\tilde{p}(0,T,x,y)\right]_{y=K} + \int_K^\infty \frac{\partial}{\partial K}(y-K)\tilde{p}(0,T,x,y)dy\right\} \\
&= e^{-rT}\left\{\cancel{\left[-(K-K)\tilde{p}(0,T,x,K)\right]} + \int_K^\infty \left(\frac{\partial}{\partial K}(y-K)\right)\tilde{p}(T,y)dy\right\} \\
&= e^{-rT}\int_K^\infty (-1)\tilde{p}(0,T,x,y)dy \\
&= -e^{-rT}\int_K^\infty \tilde{p}(0,T,x,y)dy.
\end{aligned}$$

We see that, increasing the strike makes a call option less valuable since a small increase $dK > 0$ reduces the payoff by approximately dK in every state where the call finishes in-the-money meaning that $dC \approx -e^{-rT}\mathbb{P}^{\mathbb{Q}}(S_T > K)dK$. Thus, the magnitude of C_K is the discounted probability that the option finishes in-the-money while the negative sign reflects that a higher strike lowers the call price.

Differentiating C_K again gives us convexity in strike,

$$\begin{aligned}
C_{KK} &= \frac{\partial}{\partial K}C_K(0,T,x,K) \\
&= \frac{\partial}{\partial K}\left[-e^{-rT}\int_K^\infty \tilde{p}(0,T,x,y)dy\right] \\
&= -e^{-rT}\frac{\partial}{\partial K}\int_K^\infty \tilde{p}(0,T,x,y)dy \\
&= -e^{-rT}\left[\left[-\tilde{p}(0,T,x,y)\right]_{y=K}\right] \\
&= -e^{-rT}(-\tilde{p}(0,T,x,K)) \\
&= e^{-rT}\tilde{p}(0,T,x,K).
\end{aligned}$$

As K increases, the probability $\mathbb{P}^{\mathbb{Q}}(S_T > K)$ meaning that C_K becomes less negative as K increases implying that $C_{KK} \geq 0$. More concretely, C_K is the negative discounted tail probability, while C_{KK} measures how quickly that tail probability decreases. The curvature of the call-price curve at strike K equals the discounted risk-neutral density at K. This is also known as the *Breeden-Litzenberger* identity.

More concretely we can write this as an approximation,

$$\begin{aligned} -C_K(K,T) + C_K(K+\Delta K, T) &= e^{-rT}\left[\mathbb{P}(S_T > K) - \mathbb{P}(S_T > K + \Delta K)\right] \\ &= e^{-rT}\mathbb{P}(K < S_T \leq K + \Delta K). \end{aligned}$$

Letting $\Delta K \to 0$ we get the *Breeden-Litzenberger identity*,

$$\lim_{\Delta K \to 0} e^{-rT}\mathbb{P}(K \leq S_T \leq K + \Delta K) = e^{-rT}\mathbb{P}^{\mathbb{Q}}(S_K > K) = e^{-rT}\tilde{p}(K).$$

C_T is a bit more involved in terms of computation but uses the same techniques.

Integrating by parts we get,

$$\begin{aligned} C_T(0,T,x,K) &= \frac{\partial}{\partial T}\left[e^{-rT}\int_K^\infty (y-K)\tilde{p}(0,T,x,y)\,dy\right] \\ &= -re^{-rT}\int_K^\infty (y-K)\tilde{p}(0,T,x,y)\,dy + e^{-rT}\int_K^\infty (y-K)\frac{\partial}{\partial T}\tilde{p}(0,T,x,y)\,dy \\ &= -re^{-rT}\int_K^\infty (y-K)\tilde{p}(0,T,x,y)\,dy \\ &\quad + e^{-rT}\int_K^\infty (y-K)\left[-\frac{\partial}{\partial y}\left[(r-q)y\tilde{p}(0,T,x,y)\right] + \frac{1}{2}\frac{\partial^2}{\partial y^2}\left[\sigma_{\text{loc}}^2(T,y)y^2\tilde{p}(0,T,x,y)\right]\right]dy \\ &= -re^{-rT}\int_K^\infty (y-K)\tilde{p}(0,T,x,y)\,dy - e^{-rT}\int_K^\infty (y-K)\frac{\partial}{\partial y}\left[(r-q)y\tilde{p}(T,y)\right]dy \\ &\quad + \frac{1}{2}e^{-rT}\int_K^\infty (y-K)\frac{\partial^2}{\partial y^2}\left[\sigma_{\text{loc}}^2(T,y)y^2\tilde{p}(T,y)\right]dy. \end{aligned}$$

Integrating the inner integrals by parts again gives some convenient cancellations by bounds,

$$\begin{aligned} C_T &= -re^{-rT}\int_K^\infty (y-K)\tilde{p}(T,y)dy \\ &\quad - e^{-rT}\left[(y-K)(r-q)y\tilde{p}(T,y)\right]_K^\infty + e^{-rT}\int_K^\infty (r-q)y\tilde{p}(T,y)dy \\ &\quad + \frac{1}{2}e^{-rT}\left[(y-K)\frac{\partial}{\partial y}\left[\sigma_{\text{loc}}^2(T,y)y^2\tilde{p}(T,y)\right]\right]_K^\infty \\ &\quad - \frac{1}{2}e^{-rT}\int_K^\infty \frac{\partial}{\partial y}\left[\sigma_{\text{loc}}^2(T,y)y^2\tilde{p}(T,y)\right]dy \\ &= -re^{-rT}\int_K^\infty (y-K)\tilde{p}(T,y)dy + (r-q)e^{-rT}\int_K^\infty y\tilde{p}(T,y)dy \\ &\quad - \frac{1}{2}e^{-rT}\left[\sigma_{\text{loc}}^2(T,y)y^2\tilde{p}(T,y)\right]_K^\infty \\ &= -re^{-rT}\int_K^\infty (y-K)\tilde{p}(T,y)dy + (r-q)e^{-rT}\int_K^\infty y\tilde{p}(T,y)dy \\ &\quad - \frac{1}{2}e^{-rT}\left[0 - \sigma_{\text{loc}}^2(T,K)K^2\tilde{p}(0,T,x,K)\right] \\ &= -re^{-rT}\int_K^\infty (y-K)\tilde{p}(T,y)dy + (r-q)e^{-rT}\int_K^\infty y\tilde{p}(T,y)dy \\ &\quad + \frac{1}{2}\sigma_{\text{loc}}^2(T,K)K^2e^{-rT}\tilde{p}(0,T,x,K). \end{aligned}$$

We can re-write $y := (y - K) + K$,

$$\begin{aligned} C_T &= -re^{-rT}\int_K^\infty (y-K)\tilde{p}(T,y)dy + (r-q)e^{-rT}\int_K^\infty \left[(y-K)+K\right]\tilde{p}(T,y)dy \\ &= -re^{-rT}\int_K^\infty (y-K)\tilde{p}(T,y)dy + (r-q)e^{-rT}\int_K^\infty (y-K)\tilde{p}(T,y)dy \\ &\quad + (r-q)Ke^{-rT}\int_K^\infty \tilde{p}(T,y)dy + \frac{1}{2}\sigma^2_{\text{loc}}(T,K)K^2e^{-rT}p(0,T,x,K). \end{aligned}$$

Substituting our definitions of the call-price derivatives we get,

$$\begin{aligned} C_T &= -qe^{-rT}\int_K^\infty (y-K)\tilde{p}(T,y)dy + (r-q)Ke^{-rT}\int_K^\infty \tilde{p}(T,y)dy \\ &\quad + \frac{1}{2}\sigma^2_{\text{loc}}(T,K)K^2e^{-rT}\tilde{p}(0,T,x,K) \\ &= -qC - (r-q)KC_K + \frac{1}{2}\sigma^2_{\text{loc}}(T,K)K^2C_{KK}. \end{aligned}$$

Re-arranging for $\sigma^2_{\text{loc}}(T,K)$ gives us *Dupire's Local Volatility*,

$$\sigma^2_{\text{loc}}(T,K) = \frac{2(C_T + qC + (r-q)KC_{KK})}{K^2C_{KK}}.$$

2.3 No-Arbitrage Conditions

In this section, I aim to identify where Dupire's formulation is well-defined and arbitrage-free. Well-defined and arbitrage-free in the option's volatility modelling space often go hand in hand.

Previously we had derived the local volatility formula as,

$$\sigma_{\text{loc}}^2 = \frac{2[c_T + qc + (r-q)Kc_K]}{K^2 c_{KK}}.$$

Trivially, we must have that $c_{KK} > 0$ for all strikes. But the less obvious arbitrage statements we wish to justify is why the denominator and numerator must be positive.

2.3.1 Strike-Arbitrage

Trivially, since $K^2 > 0$ we require that $c_{KK}(T,K) \geq 0$. Consequently, this says that the call price must be convex as a function of the strike. We show this by approximating the second derivative by the symmetric finite difference given by the limit,

$$\frac{\partial^2 c}{\partial K^2}(K,T) = \lim_{\varepsilon \to 0} \frac{c(K-\varepsilon,T) - 2c(K,T) + c(K+\varepsilon,T)}{\varepsilon^2}.$$

The numerator is clearly the price of the portfolio given by,

$$\frac{1}{\varepsilon^2}C(K-\varepsilon,T) - \frac{2}{\varepsilon^2}C(K,T) + \frac{1}{\varepsilon^2}C(K+\varepsilon,T).$$

Thus, we'll write the terminal payoff as,

$$B_\varepsilon(S_T) = \frac{1}{\varepsilon^2}\left[(S_T - (K-\varepsilon))^+ - 2(S_T - K)^+ + (S_T - (K+\varepsilon))^+\right].$$

I'll evaluate $B_\varepsilon(S_T)$ on each of the 4 regions it is defined on.

If $S_T \leq K - \varepsilon$, all three calls are OTM so,

$$B_\varepsilon(S_T) = 0.$$

If $K - \varepsilon < S_T \leq K$, only the $K - \varepsilon$ call is ITM so,

$$B_\varepsilon(S_T) = \frac{1}{\varepsilon^2}[S_T - K + \varepsilon] = \frac{S_T - K + \varepsilon}{\varepsilon^2}.$$

This is clearly a positive linearly rising function from 0 to $\frac{1}{\varepsilon}$.

If $S_T \geq K + \varepsilon$, all three calls are ITM so,

$$\begin{aligned} B_\varepsilon(S_T) &= \frac{1}{\varepsilon^2}\left[(S_T - K + \varepsilon) - 2(S_T - K) + (S_T - K - \varepsilon)\right] \\ &= \frac{1}{\varepsilon^2}[S_T - K + \varepsilon - 2S_T + 2K + S_T - K - \varepsilon] \\ &= 0. \end{aligned}$$

It follows that the terminal payoff function is a triangular spike centered at K,

$$B_\varepsilon(S_T) = \begin{cases} 0, & S_T \leq K - \varepsilon, \\ \dfrac{S_T - K + \varepsilon}{\varepsilon^2}, & K - \varepsilon < S_T \leq K, \\ \dfrac{K + \varepsilon - S_T}{\varepsilon^2}, & K < S_T < K + \varepsilon, \\ 0, & S_T \geq K + \varepsilon. \end{cases}$$

It has height $\frac{\varepsilon}{\varepsilon^2} = \frac{1}{\varepsilon^2}$ with base width 2ε and total area equal to $\frac{1}{2}(2\varepsilon) \cdot \frac{1}{\varepsilon} = 1$.

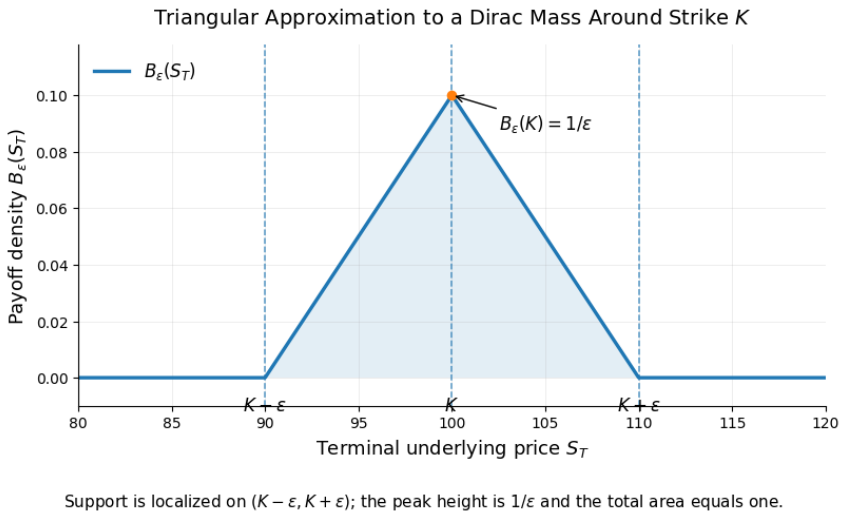


Figure 1: Triangular approximation to a Dirac mass around the strike K.

As $\varepsilon \to 0$ this payoff behaves like a Dirac delta centered at K. Moreover, it is never negative, meaning that the option price can never be negative. Therefore we conclude that

$$\frac{c(K-\varepsilon,T) - 2c(K,T) + c(K+\varepsilon,T)}{\varepsilon^2} \geq 0,$$

and taking thre limit gives that $c_{KK}(T,K) \geq 0$. We can also see this from the density,

$$c(T,K) = e^{-rT}\mathbb{E}[(S_T - K)^+].$$

Taking the first derivative with respect to strike we have,

$$\begin{aligned} c_K(T,K) &= e^{-rT}\mathbb{E}\left[\frac{\partial}{\partial K}(S_T-K)^+\right] \\ &= e^{-rT}\mathbb{E}[-\mathbf{1}\{S_T > K\}] \\ &= -e^{-rT}\mathbb{Q}(S_T > K). \end{aligned}$$

Taking the derivative of the first derivative wrt to K again,

$$\begin{aligned} c_{KK}(T,K) &= -e^{-rT}\frac{\partial}{\partial K}\mathbb{Q}(S_T > K) \\ &= -e^{-rT}\frac{\partial}{\partial K}\int_K^\infty p_T(y)dy \\ &= -e^{-rT}(-p_T(K)) = e^{-rT}p_T(K) \geq 0. \end{aligned}$$

Clearly, c_{KK} is the discounted risk-neutral density at strike K. So, if $c_{KK} < 0$, the market is assigning a negative density to some terminal stock level which is impossible and creates butterfly arbitrage.

2.3.2 Maturity Arbitrage

The numerator of our local volatility equation contains the component $c_T + qc + (r-q)Kc_K$. To see the maturity-arbitrage condition more clearly, we can re-write this as a maturity derivative of a forward-moneyness-normalized call-option price. Recall that the forward price at time 0 to maturity T with fixed proportional strike for $k > 0$ is written as $F_T = xe^{(r-q)T}$. The actual strike at maturity T is written as,

$$K(T) = kF_T.$$

We also define $h(T) = e^{qT}c(T, K(T))$ and differentiate $h(T)$ using the product and chain rule,

$$\begin{aligned}
h'(T) &= \frac{d}{dT}\left[e^{qT}c(T, K(T))\right] \\
&= qe^{qT}c(T, K(T)) + e^{qT}\frac{d}{dT}c(T, K(T)) \\
&= qe^{qT}c(T, K(T)) + e^{qT}\left[c_T(T, K(T)) + c_K(T, K(T))K'(T)\right] \\
&= e^{qT}\left[qc(T, K(T)) + c_T(T, K(T)) + c_K(T, K(T))K'(T)\right] \\
&= e^{qT}\left[qc + c_T + (r-q)Kc_K\right].
\end{aligned}$$

This follows because $K(T) = kF_T = kxe^{(r-q)T}$ so $K'(T) = (r-q)kxe^{(r-q)T} = (r-q)K(T)$ by definition. Hence, the numerator is positive if and only if $T \mapsto e^{qT}c(T, kF_T)$ is increasing for each fixed strike k. Explicitly,

$$c_T + qc + (r-q)Kc_K = e^{-qT}\frac{d}{dT}\left[e^{qT}c(T, kF_T)\right].$$

In the forward pricing sense, if we compare options at the same forward moneyness that is $k = K/F_T$, then the appropriately adjusted call price should increase with maturity. For $T_1 < T_2$ the no-arbitrage condition is,

$$e^{qT_1}c(T_1, kF_{T_1}) \leq e^{qT_2}c(T_2, kF_{T_2}).$$

Suppose for a second that this condition is violated. Then that means there exists $T_1 < T_2$ and k such that,

$$e^{qT_1}c(T_1, kF_{T_1}) > e^{qT_2}c(T_2, kF_{T_2}) \iff c(T_2, kF_{T_2}) < e^{-q(T_2-T_1)}c(T_1, kF_{T_1}).$$

So if we buy 1 call with maturity T_2 and strike kF_{T_2}, we incure a negative initial cost meaning we receive money at inception. By selling $e^{-q(T_2-T_1)}$ calls with maturity T_1 and strike kF_{T_1} gives an initial cost of,

$$c(T_2, kF_{T_2}) - e^{-q(T_2-T_1)}c(T_1, kF_{T_1}) < 0,$$

which is negative by the violated inequality. Thus our portfolio is constructed by being long a call with maturity T_2 and short a call with maturity T_1. The payoffs of the long and short maturity calls are respectively,

$$(S_{T_2} - kF_{T_2})^+, \qquad e^{-q\Delta T}(S_{T_1} - kF_{T_1})^+.$$

We must finance to time T_2 so the payoff becomes,

$$e^{r\Delta T}e^{-q\Delta T}(S_{T_1} - kF_{T_1})^+ = e^{(r-q)\Delta T}(S_{T_1} - kF_{T_1})^+.$$

In terms of forward pricing, the financing of the payoff above can be written as,

$$\frac{F_{T_2}}{F_{T_1}} = \frac{xe^{(r-q)T_2}}{xe^{(r-q)T_1}} = e^{(r-q)(T_2-T_1)} = e^{(r-q)\Delta T}.$$

We also enter a delta position in the stock, that is, $\Delta = 0$ if the call option is OTM and (-1) if the option is ITM. Therefore, the self-financing PnL from holding Δ shares from T_1 to T_2 after financing and dividends becomes,

$$\Delta\left\{\frac{F_{T_2}}{F_{T_1}}(S_{T_1} - kF_{T_1})^+\right\} = \Delta\left(S_{T_2} - \frac{F_{T_2}}{F_{T_1}}S_{T_1}\right).$$

Therefore the total PnL at time T_2 is given by,

$$\Pi_{T_2} = (S_{T_2} - kF_{T_2})^+ - \frac{F_{T_2}}{F_{T_1}}(S_{T_1} - kF_{T_1})^+ + \Delta\left(S_{T_2} - \frac{F_{T_2}}{F_{T_1}}S_{T_1}\right).$$

Suppose we also have a forward-scaled T_1 stock valued at $S^*_{T_1} = \frac{F_{T_2}}{F_{T_1}}S_{T_1}$ which means that $\frac{F_{T_2}}{F_{T_1}}kF_{T_1} = kF_{T_2}$. So naturally, the payoff of the financed payoff of the short-term maturity T_1 call is,

$$\frac{F_{T_2}}{F_{T_1}}(S_{T_1} - kF_{T_1})^+ = (S^*_{T_1} - kF_{T_2})^+.$$

So it is the case that $S_{T_1} > kF_{T_1}$ is equivalent to $S^*_{T_1} > kF_{T_2}$.

Therefore our delta is $-\mathbf{1}\{S^*_{T_1} > kF_{T_2}\}$ so the PnL of our position is,

$$\Pi_{T_2} = \underbrace{(S_{T_2} - kF_{T_2})^+}_{=f(S_{T_2})} - \underbrace{(S^*_{T_1} - kF_{T_2})^+}_{=f(S^*_{T_1})} - \underbrace{\mathbf{1}\{S^*_{T_1} kF_{T_2}\}}_{=f'(S^*_{T_1})}(S_{T_2} - S^*_{T_1}),$$

if we let $f(x) = (x - kF_{T_2})^+$. Clearly this is a convex function, by definition. That is, for any convex function, $f(y) \geq f(x) + f'(x)(y - x)$ by *Supporting-Tangent Inequality*. Thus we get that,

$$f(S_{T_2}) \geq f(S^*_{T_2}) + f'(S^*_{T_1})(S_{T_2} - S^*_{T_1}).$$

If the maturity monotonicity condition is violated, we receive money at inception and still end with a non-negative terminal PnL. This is clearly maturity arbitrage.

More specifically, under risk-neutral measure with dividend yield,

$$\mathbb{E}^{\mathbb{Q}}[S_{T_2} \mid S_{T_1}] = S_{T_1} e^{(r-q)(T_2-T_1)} = \frac{F_{T_2}}{F_{T_1}} S_{T_1}.$$

Defining $f(x) = (x - kF_{T_2})^+$, by Jensen's inequality, since f is convex, we have that,

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}[f(S_{T_2}) \mid S_{T_1}] &\geq f\left(\mathbb{E}^{\mathbb{Q}}[S_{T_2} \mid S_{T_1}]\right) \\
&= f\left(\frac{F_{T_2}}{F_{T_1}} S_{T_1}\right) \\
&= \left(\frac{F_{T_2}}{F_{T_1}} - kF_{T_2}\right)^+ \\
&= \left(\frac{F_{T_2}}{F_{T_1}}(S_{T_1} - kF_{T_1})\right)^+ \\
&= \frac{F_{T_2}}{F_{T_1}}(S_{T_1} - kF_{T_2})^+.
\end{aligned}$$

Taking unconditional expectations we get,

$$\mathbb{E}^{\mathbb{Q}}\left[(S_{T_2} - kF_{T_2})^+\right] \geq \frac{F_{T_2}}{F_{T_1}} \mathbb{E}^{\mathbb{Q}}\left[(S_{T_1} - kF_{T_1})^+\right].$$

Multiplying both sides by $e^{-(r-q)T_2}$ we see that,

$$\begin{aligned}
e^{-(r-q)T_2}\mathbb{E}^{\mathbb{Q}}\left[(S_{T_2} - kF_{T_2})^+\right] &\geq e^{-(r-q)T_2} e^{(r-q)(T_2-T_1)} \mathbb{E}^{\mathbb{Q}}\left[(S_{T_1} - kF_{T_1})^+\right] \\
&= e^{-(r-q)T_1} \mathbb{E}^{\mathbb{Q}}\left[(S_{T_1} - kF_{T_1})^+\right].
\end{aligned}$$

However, observe that,

$$\begin{aligned}
e^{qT} c(T, kF_T) &= e^{qT} e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - kF_T)^+] \\
&= e^{-(r-q)T} \mathbb{E}^{\mathbb{Q}}\left[(S_T - kF_T)^+\right].
\end{aligned}$$

Therefore it must be the case that,

$$e^{qT_2} c(T_2, kF_{T_2}) \geq e^{qT_1} c(T_1, kF_{T_1}).$$

Therefore, in any arbitrage-free model, the maturity-adjusted price at fixed forward moneyness must increase with maturity. Therefore, the full no-arbitrage interpretation of Dupire's local volatility model is only well-defined when $c_{KK} \geq 0$ and when $c_T + qc + (r - q)Kc_K \geq 0$.

A butterfly spread cannot have negative price and non-negative payoff. Moreover, adjusted call prices must increase with maturity given a fixed forward-moneyness because then we can simply buy the longer option and sell the shorter option, delta-adjust the short maturity and lock in an arbitrage profit.

2.3.3 Convex-Order Condition for Implied Volatilities

Suppose again we fixed forward-moneyness, that is fix $K = kF_T$ for $F_T = S_0 e^{(r-q)T}$.

Let us also define *total implied variance* as,

$$\Sigma(K,T) = \sigma_{\text{imp}}(K,T)\Big|_{K=kF_T}, \quad w(K,T) = T\Sigma^2(K,T).$$

It follows from the Black-Scholes formula for a call option price that,

$$\begin{aligned} c(T,K) &= S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2) \\ &= S_0 e^{-qT} N(d_1) - k S_0 e^{(r-q)T} e^{-rT} N(d_2), \qquad K = kF_T = kS_0 e^{(r-q)T} \\ &= S_0 e^{-qT} \left\{ N(d_1) - K N(d_2) \right\}. \end{aligned}$$

Multiplying both sides by e^{qT} we get,

$$\begin{aligned} e^{qT} c(T,K) &= S_0 \left[N(d_1) - K N(d_2) \right] \\ e^{qT} c(T,kF_T) &= S_0 B(k, w(K,T)) \\ &:= S_0 \left\{ N\left(\frac{-\log k + \frac{1}{2}w}{\sqrt{w}} \right) - K N\left(\frac{-\log k - \frac{1}{2}w}{\sqrt{w}} \right) \right\}. \end{aligned}$$

We'll define,

$$B(k, w(K,T)) := \left\{ N\left(\frac{-\log k + \frac{1}{2}w}{\sqrt{w}} \right) - K N\left(\frac{-\log k - \frac{1}{2}w}{\sqrt{w}} \right) \right\}.$$

What is left is to express the factors d_1 and d_2 in terms of total implied variance.

The algebra follows directly,

$$\begin{aligned} d_1 &= \frac{\log\left(\frac{S_0}{K}\right) + \left(r - q + \frac{1}{2}\Sigma^2\right) T}{\Sigma\sqrt{T}} \\ &= \frac{-\log K + \frac{1}{2}\Sigma^2 T}{\Sigma\sqrt{T}} \\ &= \frac{-\log K + \frac{1}{2}w}{\sqrt{w}}. \end{aligned}$$

Note that the moneyness term in the numerator simplies after substituting,

$$\log \frac{S_0}{K} + (r-q)T = \log\left\{ \frac{S_0}{K S_0 (r-q)T} \right\} + (r-q)T = -\log K - (r-q)T + (r-q)T = -\log K.$$

Then, d_2 follows through the identity,

$$d_1 = d_2 + \Sigma^2\sqrt{T} \implies d_2 = d_1 - \Sigma^2\sqrt{w} = \frac{-\log k - \frac{1}{2}w}{\sqrt{w}}.$$

Therefore the convexity condition on implied variances is equivalent,

$$\begin{aligned} e^{qT_2} c(T_2, KF_{T_2}) \geq e^{qT_1} c(T_2, KF_{T_1}) &\iff S_0 B(k, w(K,T_2)) \geq S_0 B(k, w(K,T_1)) \\ &\iff B(k, w(K,T_2)) \geq B(k, w(K,T_1)). \end{aligned}$$

Let $m = \log k$ then,

$$d_1 - d_2 = \sqrt{w}, \quad d_1 + d_2 = -\frac{2m}{\sqrt{w}}.$$

The difference in squares of the factors is therefore,

$$d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2) = \sqrt{w} \cdot \left(-\frac{2m}{\sqrt{w}}\right) = -2m.$$

Using the normal cdf we get that,

$$\frac{\phi(d_1)}{\phi(d_2)} = \frac{e^{-d_1^2/2}}{e^{-d_2^2/2}} = e^{-(-2m)/2} = e^m = k.$$

Taking the derivative of B wrt w we get,

$$\begin{aligned}
\frac{\partial B}{\partial w} &= \phi(d_1) \cdot \frac{\partial d_1}{\partial w} - K\phi(d_2) \cdot \frac{\partial d_2}{\partial w} \\
&= \phi(d_1) \cdot \left(\frac{\partial d_1}{\partial w} - \frac{\partial d_2}{\partial w}\right) \\
&= \phi(d_1) \cdot \frac{\partial}{\partial w} \cdot (d_1 - d_2) \\
&= \phi(d_1) \cdot \frac{\partial}{\partial w} \sqrt{w} \\
&= \frac{1}{2\sqrt{w}} \cdot \phi(d_1),
\end{aligned}$$

which is always positive since $w > 0$ and $\phi(d_1) > 0$.

We conclude that $B(K, w)$ is a strictly increasing function in total variance.

More explicitly, it follows that,

$$\begin{aligned}
B(K, w(K, T_2)) &\geq B(K, w(K, T_1)) \\
w(K, T_2) &= w(K, T_1) \\
T_2\Sigma^2(K, T_2) &\geq T_1\Sigma^2(K, T_1).
\end{aligned}$$

Solving for $\Sigma(K, T_2)$ we get, that,

$$\Sigma(K, T)_2 \geq \sqrt{\frac{T_1}{T_2}}\Sigma^2(K, T_1).$$

2.4 Implied Volatilities to Local Volatilities

As we saw previously, it is often easier to view the strike in terms of forward log-moneyness, that is, to write the strike explicitly as a forward price level. We can therefore express K as $K = kF_T = kS_0e^{(r-q)T}$.

Therefore, the call price under Black-Scholes can be written as,

$$\begin{aligned}
C(T, K) &= S_0e^{-qT}N(d_1) - Ke^{-rT}N(d_2) \\
&= S_0e^{-qT}N(d_1) - KS_0e^{(r-q)T}e^{-rT}N(d_2) \\
&= S_0\left\{e^{-qT}N(d_1) - Ke^{(r-q)T}e^{-rT}N(d_2)\right\} \\
&= S_0e^{-qT}\left\{N(d_1) - KN(d_2)\right\}.
\end{aligned}$$

We'll let the the initial price term involving the exponent be $A(T)$ and let the difference of the normal CDF be a function of log forward-moneyness, that is $B(y, f(T, y))$. $f(T, y)$ will be defined as the total implied variance, that is $f(T, y) = T\hat{\sigma}^2(K, T)$ and $A(T) = S_0e^{-qT}$ and $B(y, f(T, y)) = N(d_1) - KN(d_2)$.

Recall that Dupire's model of Local Volatility can be written as,

$$\sigma_{\text{loc}}^2 = \frac{2[c_T + qc + (r - q)Kc_K]}{K^2c_{KK}}.$$

Thus we solve explicitly for the derivatives c_T, c_K, c_{KK}. The derivatives of y are as follows,

$$\frac{\partial y}{\partial y} = -(r-q), \quad \frac{\partial y}{\partial K} = \frac{1}{K},$$

for a fixed strike K and fixed maturity T. c_T follows directly from the chain rule,

$$\begin{aligned} c_T &= A_T B + A\left[B_y y_T + B_f(f_T + f_y y_T)\right] \\ &= -qAB + A\left\{-(r-q)B_y + B_f f_T - (r-q)B_f f_y\right\} \\ &= A\left\{-qB + B_f f_T - (r-q)\left(B_f + B_f f_y\right)\right\}. \end{aligned}$$

We will keep this formulae for reference. c_K, again, follows directly from an application of chain rule,

$$\begin{aligned} c_K &= A\left\{B_y + B_f f_y\right\} \cdot y_K \\ &= A\left\{B_y + B_f f_y\right\} \frac{1}{K} \\ Kc_K &= A\left\{B_y + B_f f_y\right\} \\ \iff (r-q)Kc_K &= (r-q)A\left\{B_y + B_f f_y\right\}. \end{aligned}$$

Substituting these derivatives back into the numerator of Dupire's local volatility,

$$\begin{aligned} c_T + qC + (r-q)Kc_K &= A\left\{\cancel{-qB} + B_f f_T - \cancel{(r-q)(B_y + B_f f_y)}\right\} + \cancel{qAB} + \cancel{(r-q)A(B_y + B_f f_y)} \\ &= AB_f f_T \\ &= S_0 e^{-qT} B_T f_T. \end{aligned}$$

For algebraic convenience, we will write $G := B_y + B_f f_y$ so that c_K becomes $c_K = A \cdot \frac{G}{K}$.

Then $c_K K$ follows directly from chain rule,

$$\begin{aligned} c_{KK} &= A \cdot \frac{\partial}{\partial K}\left(\frac{G}{K}\right) \\ &= A\left\{\frac{G_y}{K^2} - \frac{G}{K^2}\right\} \\ &= \frac{A}{K^2}\left\{G_y - G\right\} \\ \implies K^2 c_{KK} &= A\left\{G_y - G\right\}. \end{aligned}$$

G_y follows from another application of chain rule,

$$\begin{aligned} G_y &= B_{yy} + \frac{\partial}{\partial y}(B_f f_y) \\ &= B_{yy} + B_{yf} f_y + B_{ff} f_y^2 + B_f f_{yy} + B_{fy} f_y \\ &= B_{yy} + 2B_{yf} f_y + B_{ff} f_y^2 + B_f f_{yy}. \end{aligned}$$

The derivatives B_y and B_{yy} follow as,

$$\begin{aligned}
B_y &= \phi(d_1)d_{1,y} - e^y N(d_2) - e^y\phi(d_2)d_{2,y} \\
&= -\frac{\phi(d_1)}{\sqrt{f}} - e^y N(d_2) + \frac{e^y\phi(d_2)}{\sqrt{f}} \\
&= \phi(d_1)d_{1,f} - e^y\phi(d_2)d_{2,f} \\
&= \phi(d_1)(d_{1,f} - d_{2,f}) \\
&= \frac{1}{2\sqrt{f}}\phi(d_1) \\
B_{yy} &= -e^y N(d_2) - e^y\phi(d_2)d_{2,y} \\
&= B_y - e^y\phi(d_2)\left(-\frac{1}{\sqrt{f}}\right) \\
&= B_y + \frac{e^y\phi(d_2)}{\sqrt{f}} \\
&= B_y + \phi(d_1)\cdot\frac{1}{\sqrt{f}} \\
&= B_y + 2B_f.
\end{aligned}$$

Therefore $B_{yy} - B_y = 2B_f$. We will require the differences,

$$d_2 = -yf^{1/2} - \frac{1}{2}f^{1/2}, \quad d_{2,f} = \frac{y}{2}f^{-3/2} - \frac{1}{4}f^{-1/2} = \frac{y}{2y^{3/2}} - \frac{1}{4\sqrt{f}}.$$

Then it follows that B_{yf} is,

$$\begin{aligned}
B_{yf} &= -\phi(d_1)\left\{\frac{y}{2f^{3/2}} - \frac{1}{4\sqrt{f}}\right\} \\
&= \phi(d_1)\left\{\frac{1}{4\sqrt{f}} - \frac{y}{2f^{3/2}}\right\} \\
&= \frac{\phi(d_1)}{2\sqrt{f}}\left\{\frac{1}{2} - \frac{y}{f}\right\} \\
&= B_f\left\{\frac{1}{2} - \frac{y}{f}\right\}.
\end{aligned}$$

Therefore the difference term becomes,

$$2B_{yf} - B_F = 2B_f\left\{\frac{1}{2} - \frac{y}{f}\right\} - B_y = B_f - \frac{2y}{f}B_f - B_f = -\frac{2y}{f}B_f.$$

What remains is to determine what B_{ff} is in closed-form.

Recall that we had that $B_f = \frac{\phi(d_1)}{2\sqrt{f}}$ and we can write the log of B_f as,

$$\log B_f = \log\phi(d_1) - \log 2 - \frac{1}{2}\log f.$$

To simplify the cancellations a bit we will write the derivatives as a ratio,

$$\begin{aligned}
\frac{B_{ff}}{B_f} &= \frac{\partial}{\partial f}\log B_f \\
&= \frac{\partial}{\partial f}\log\phi(d_1) - \frac{1}{2f} \\
&= -d_1 d_{1,f} - \frac{1}{2f} \\
&= -\left\{-\frac{y}{\sqrt{f}} + \frac{1}{2}\sqrt{f}\right\}\cdot\left\{\frac{y}{2f^{3/2}} + \frac{1}{4\sqrt{f}}\right\} - \frac{1}{2f} \\
&= -\left(\frac{y^2}{2f^2} - \cancel{\frac{y}{4f}} + \cancel{\frac{y}{4f}} + \frac{1}{8}\right) - \frac{1}{2f} \\
&= -\left\{-\frac{y^2}{2f^2} + \frac{1}{8}\right\} - \frac{1}{2f} \\
&= \frac{y^2}{2f^2} - \frac{1}{2f} - \frac{1}{8} \\
&= \frac{1}{2}\left(\frac{y^2}{f^2} - \frac{1}{f} - \frac{1}{4}\right).
\end{aligned}$$

Substituting this into the difference $G_y - G$,

$$\begin{aligned}
G_y - G &= 2B_f - \frac{2y}{f}B_f f_y + \frac{1}{2}B_f\left\{\frac{y^2}{f^2} - \frac{1}{f} - \frac{1}{4}\right\}f_y^2 + B_f f_{yy} \\
&= 2B_f\left\{1 - \frac{y}{f}f_y + \frac{1}{4}\left(\frac{y^2}{f^2} - \frac{1}{f} - \frac{1}{4}\right)f_y^2 + \frac{1}{2}f_{yy}\right\}.
\end{aligned}$$

Re-arranging for squared strike scaled by convexity with respect to strike we get,

$$K^2 c_{KK}. = S_0 e^{-qT}\cdot 2B_f\left\{1 - \frac{y}{f}f_y + \frac{1}{2}f_{yy} + \frac{1}{4}\left(\frac{y^2}{f^2} - \frac{1}{f} - \frac{1}{4}f_y^2\right)\right\}$$

Substituting our definitions we get,

$$\begin{aligned}
\sigma^2_{\text{loc}}(T,K) &= \frac{2S_0 e^{-qT}B_f f_T}{S_0 e^{-qT}\cdot 2B_f\left[1 - \frac{y}{f}f_y + \frac{1}{2}f_{yy} + \frac{1}{4}\left(\frac{y^2}{f^2} - \frac{1}{f} - \frac{1}{4}\right)f_y^2\right]} \\
&= \frac{f_T}{1 - \frac{y}{f}f_y + \frac{1}{2}f_{yy} + \frac{1}{4}\left(\frac{y^2}{f^2} - \frac{1}{f} - \frac{1}{4}\right)f_y^2} \\
&= \frac{f_T}{1 - \frac{y}{f}f_y + \frac{y^2}{4f^2}f_y^2 + \frac{1}{2}f_{yy} - \frac{1}{4}\left(\frac{1}{f} + \frac{1}{4}\right)f_y^2} \\
&= \left.\frac{f_T}{\left(1 - \frac{y}{2f}f_y\right)^2 + \frac{1}{2}f_{yy} - \frac{1}{4}\left(\frac{1}{f} + \frac{1}{4}\right)f_y^2}\right|_{y=\log(K/F_T)}.
\end{aligned}$$

This formulation provides us with a direct mapping from the implied total variance surface to the local volatility surface. The numerator, in particular, is the maturity slope of total implied variance at a fixed forward log-moneyness, or in other words, the maturity-arbitrage condition where $f_T(T,y) \geq 0$. The denominator, evidently, is proportional to c_{KK} because $K^2 c_{KK} = S_0 e^{-qT}\cdot 2B_f$ scaled by some denominator.

Since $S_0 e^{-qT} > 0$ and $B_f > 0$, the denominator positive exactly when the call surface is convex in strike, that is, when there is no butterfly arbitrage. So, the implied-to-local vol formula is really just a Dupire local volatility model re-written in the coordinate (y,T,f) that is $(K,T,\hat{\sigma}) \mapsto (y,T,f)$ where $y = \log(K/F_T)$ and $f = T\hat{\sigma}^2$. Consider the flat volatility case as a sanity check.

Suppose that $\hat{\sigma}(K,T) = \sigma_0$ for all fixed strikes and maturities. Then $f(T,y) = \sigma_0^2 T$ so that $f_T = \sigma_0^2$ and $f_y = 0$ with $f_{yy} = 0$. The denominator is trivially just 1 so $\sigma^2_{\text{loc}}(T,S) = \sigma_0^2$.

Therefore, the flat implied volatility surface produces a flat local volatility surface, as it should.

In practice, we take market implied volatilities, convert them to total variances $f = T\hat{\sigma}^2$, express them as a smooth surface (T,y) and compute f_T, f_y, f_{yy} and the plug those derivatives into the formulation above. Empirically, this is the reason why the interpolation of the implied volatility surface matters, that is, the local volatility is not just using implied volatility levels but rather using their first maturity derviative, first moneyness derivative and second moneyness derivative.

2.5 Interpreting Implied Variances through Local Volatilities

Consider the following stock-price dynamics SDE where $\sigma_1(t,S_t)$ is Black-Scholes constant volatility,

$$dS_t^{(1)} = (r-q)S_t dt + \sigma_1(t,S_t)S_t, dW_t^{(1)}.$$

Note that this SDE satisfies the PDE,

$$0 = P_{1,t} + (r-q)SP_{1,S} + \frac{1}{2}\sigma_1^2(t,S)S^2 P_{1,SS} - rP_1.$$

If we re-arrange we get the dollar-weighted gamma term in the Black-Scholes PDE,

$$-\frac{1}{2}\sigma_1^2(t,S)S^2 P_{1,SS} = P_t + (r-q)SP_{1,S} - rP_1.$$

Consider another stock-price dynamics SDE, this time allowing the instantaneous variance term be arbitrary i.e., local volatility, stochastic volatility,

$$dS_t^{(2)} = (r-q)S_t dt + \sigma_{2t}S_t dW_t^{(2)}.$$

By application of Ito's lemma letting $Q_t = e^{-rt}P_1(t,S_t)$,

$$\begin{aligned}
dQ_t &= d\left(e^{-rt}P_1(t,S_t)\right) \\
&= e^{-rt}dP_1(t,S_t) - re^{-rt}P_1(t,S_t)dt \\
&= e^{-rt}\left[P_{1,t}dt + P_{1,S}dS_t^{(2)} + \frac{1}{2}P_{1,SS}d\langle S\rangle_t - rP_1 dt\right] \\
&= e^{-rt}\left[P_{1,t}dt + P_{1,S}\left((r-1)S_t dt + \sigma_{2t}S_t dW_t^{(2)}\right) + \frac{1}{2}P_{1,SS}\sigma_{2t}^2 S_t^2 dt - rP_1 dt\right],
\end{aligned}$$

where the last line follows by substituting the original stock-price dynamics. Re-arranging,

$$\begin{aligned}
&= e^{-rt}\left[-\frac{1}{2}\sigma_1^2(t,S_t)S_t^2 P_{1,SS} + \frac{1}{2}\sigma_{2t}^2 S_t^2 P_{1,SS}\right]dt + e^{-rt}\sigma_{2t}S_t P_{1,S}dW_t^{(1)} \\
&= \frac{1}{2}e^{-rt}S_t^2 P_{1,SS}\left(\sigma_{2t}^2 - \sigma_1^2(t,S_t)\right)dt + e^{-rt}\sigma_{2t}S_t P_{1,S}dW_t^{(2)}.
\end{aligned}$$

Taking expectations under the arbitrary stochastic price model dynamics,

$$\begin{aligned}
\mathbb{E}[dQ_t] &= \mathbb{E}_2\left[\frac{1}{2}e^{-rt}S_t^2 P_{1,SS}\left(\sigma_{2t}^2 - \sigma_1^2(t,S_t)\right)dt + e^{-rt}\sigma_{2t}S_t P_{1,S}dW_t^{(2)}\right] \\
&= \frac{1}{2}e^{-rt}S_t^2 P_{1,SS}(\sigma_{2t}^2 - \sigma_1^2(t,S_t))dt + \cancel{e^{-rt}\sigma_{2t}S_t P_{1,S}\mathbb{E}_2[dW_t^{(2)} \mid \mathcal{F}_t]} \\
&= \frac{1}{2}e^{-rt}S_t^2 P_{1,SS}\left(\sigma_{2t}^2 - \sigma_1^2(t,S_t)\right)dt.
\end{aligned}$$

Integrating,

$$\begin{aligned} Q_T &= Q_0 + \int_0^T dQ_t \\ &= Q_0 + \int_0^T \frac{1}{2} e^{-rt} S_t^2 P_{1,SS} \left(\sigma_{2t}^2 - \sigma_1^2(t, S_t)\right) dt \\ &\quad \int_0^T e^{-rt} \sigma_{2t} S_t P_{1,S} dW_t^{(2)}. \end{aligned}$$

Taking expectations again under the stochastic stock price dynamics model,

$$\begin{aligned} \mathbb{E}_2[Q_T] &= \mathbb{E}_2[Q_0] + \mathbb{E}_2\left[\int_0^T \frac{1}{2} e^{-rt} S_t^2 P_{1,SS} \left(\sigma_{2t}^2 - \sigma_1^2(t, S_t)\right) dt\right] \\ &\quad \mathbb{E}_2\left[\int_0^T e^{-rt} \sigma_{2t} S_t P_{1,S} dW_t^{(2)}\right] \\ &= \underbrace{Q_0}_{:=P_1(0,S_0)} + \mathbb{E}_2\left[\int_0^T \frac{1}{2} e^{-rt} S_t^2 P_{1,SS} \left(\sigma_{2t}^2 - \sigma_1^2(t, S_t)\right) dt\right]. \end{aligned}$$

Recall that under risk-neutral expectations we can write,

$$\mathbb{E}_2[Q_T] = \mathbb{E}_2\left[e^{-rt} P_1(T, S_T)\right] = \mathbb{E}_2[e^{-rt} h(S_T)] = P_2(0, S_0).$$

Therefore we can write the initial price under the stochastic model as,

$$P_2(0, S_0) = P_1(0, S_0) + \mathbb{E}_2\left[\int_0^T \frac{1}{2} e^{-rt} S_t^2 P_{1,SS}(t, S_t) \cdot (\sigma_{2t}^2 - \sigma_1^2(t, S_t)) dt\right].$$

Let us denote the price under the initial model as the Black-Scholes model, or more explicitly, with respect to $\hat{\sigma}_{KT}$ we write $P_1(0, S_0) := P_{\hat{\sigma}_{KT}}(0, S_0)$. That is, under Black-Scholes dynamics, for a specific strike K and maturity T, the respective price is $P_{\hat{\sigma}_{KT}}(0, S_0)$. To validate the model, we require that at time $t = 0$, the stochastic model price should equal the constant-variance Black-Scholes price, that is,

$$P_{\hat{\sigma}_{KT}}(0, S_0) = P_2(0, S_0).$$

Solving and simplifying we get,

$$\begin{aligned} 0 &= P_2(0, S_0) - P_{\hat{\sigma}_{KT}}(0, S_0) \\ &= \mathbb{E}_2\left[\int_0^T \frac{1}{2} e^{-rt} S_t^2 P_{1,SS}(t, S_t)(\sigma_{2t}^2 - \sigma_1^2(t, S_t)) dt\right] \\ &= \mathbb{E}_2\left[\int_0^T \frac{1}{2} e^{-rt} S_t^2 \frac{\partial^2 P_{\hat{\sigma}_{KT}}(t, S_t)}{\partial S^2}(\sigma_{2t}^2 - \sigma_1^2(t, S_t)) dt\right] \\ &= \mathbb{E}_2\left[\int_0^T \frac{1}{2} e^{-rt} S_t^2 \frac{\partial P_{\hat{\sigma}_{KT}}}{\partial S^2}(t, S_t)\sigma_{2t}^2\right] - \mathbb{E}_2\left[\int_0^T \frac{1}{2} e^{-rt} S_t^2 \frac{\partial^2 P_{\hat{\sigma}_{KT}}}{\partial S^2}(t, S_t)\hat{\sigma}_{KT}^2 dt\right] \\ &= \mathbb{E}_2\left[\int_0^T \frac{1}{2} e^{-rt} S_t^2 \frac{\partial P_{\hat{\sigma}_{KT}}}{\partial S^2}(t, S_t)\sigma_{2t}^2\right] - \hat{\sigma}_{KT}^2 \cdot \mathbb{E}_2\left[\int_0^T \frac{1}{2} e^{-rt} S_t^2 \frac{\partial^2 P_{\hat{\sigma}_{KT}}}{\partial S^2}(t, S_t) dt\right] \end{aligned}$$

Re-arranging,

$$\hat{\sigma}_{KT}^2 = \frac{\mathbb{E}_2\left[\int_0^T e^{-rt} S_t^2 \frac{\partial P_{\hat{\sigma}_{KT}}}{\partial S^2}(t, S_t)\sigma_{2t}^2 dt\right]}{\mathbb{E}_2\left[\int_0^T e^{-rt} S_t^2 \frac{\partial^2 P_{\hat{\sigma}_{KT}}}{\partial S^2}(t, S_t) dt\right]} = \frac{\mathbb{E}_2\left[\int_0^T e^{-rt} S_t^2 \Gamma_{\hat{\sigma}_{KT}}(t, S_t)\sigma_{2t}^2 dt\right]}{\mathbb{E}_2\left[\int_0^T e^{-rt} S_t^2 \Gamma_{\hat{\sigma}_{KT}}(t, S_t) dt\right]}$$

Appendix

Vanishing δt

From the derivation provided in Section 1.1 we utilized the second-order multivariable Taylor expansion of the option premium $P = P(t, S)$ where two variables are time t and spot S. In option pricing we only keep terms that survive order δt because δS is stochastic and satisfies $(\delta S)^2 \sim O(\delta t)$. Starting from the multivariable Taylor expansion around (t, S) we write,

$$P(t+\delta t, S+\delta S) = P(t,S)+P_t(t,S)\delta t+P_S(t,S)\delta S+\frac{1}{2}\begin{pmatrix}\delta t & \delta S\end{pmatrix}\begin{pmatrix}P_{tt}(t,S) & P_{tS}(t,S)\\ P_{St}(t,S) & P_{SS}(t,S)\end{pmatrix}\begin{pmatrix}\delta t\\ \delta S\end{pmatrix} \quad + \quad \text{Higher-order terms.}$$

Multiplying out the Hessian quadratic form above we get,

$$\begin{aligned}\frac{1}{2}\begin{pmatrix}\delta t & \delta S\end{pmatrix}\begin{pmatrix}P_{tt}(t,S) & P_{tS}(t,S)\\ P_{St}(t,S) & P_{SS}(t,S)\end{pmatrix}\begin{pmatrix}\delta t\\ \delta S\end{pmatrix} &= \frac{1}{2}\begin{pmatrix}\delta t & \delta S\end{pmatrix}\begin{pmatrix}P_{tt}(t,S)\delta t & P_{tS}(t,S)\delta S\\ P_{St}(t,S)\delta t & P_{SS}(t,S)\delta S\end{pmatrix}\\ &= \frac{1}{2}\left[P_{tt}(\delta t)^2 + P_{tS}\delta t\delta S + P_{St}\delta S\delta t + P_{SS}(\delta S)^2\right]\\ &= \frac{1}{2}P_{tt}(\delta t)^2 + P_{tS}\delta t\delta S + \frac{1}{2}P_{SS}(\delta S)^2.\end{aligned}$$

Therefore the full second-order Taylor expansion becomes,

$$= P(t,S) + P_t\delta t + P_S\delta S + \frac{1}{2}P_{tt}(\delta t)^2 + P_{tS}\delta t\delta S + \frac{1}{2}P_{SS}(\delta S)^2 + \cdots$$

Under Black-Scholes, the stock increment δS satisfies,

$$\delta S = \mu S\delta t + \sigma S\delta W, \qquad \delta W \sim O(\sqrt{\delta t}).$$

The dominant random side of δS is of order $O(\sqrt{\delta t})$ meaning that $(\delta S)^2 \sim O(\delta t)$, $\delta t\delta S \sim O((\delta t)^{3/2})$ and $(\delta t)^2 \sim O((\delta t)^2)$. The terms of order $(\delta t)^{3/2}$ and $(\delta t)^2$ vanish faster than δt so they are discarded in the continuous-time limit which leaves us with the incremented option premium formulation given as,

$$P(t+\delta t, S+\delta S) = P(t,S) + P_t\delta t + P_S\delta S + \frac{1}{2}P_{SS}(\delta S)^2 + o(\delta t).$$

Dropping the smaller order error gives exactly our expansion used in Section 1.1,

$$P(t+\delta t, S+\delta S) \approx P(t,S) + P_t\delta t + P_S\delta S + \frac{1}{2}P_{SS}(\delta S)^2.$$

We can show the continuous-time limit by dividing every Taylor term by δt then take the limit as $\delta t \to 0$ i.e., the limit as our time intervals become smaller and smaller. Anything whose ratio to δt goes to zero is negligible at the dt-scale. Suppose we let $h := \delta t$ and for the stock process locally we assume that $\delta S = \mu Sh + \sigma S\delta W$ where $\delta W = \sqrt{h}Z$ for $Z \sim \mathcal{N}(0,1)$. Therefore we can write $\delta S = \mu Sh + \sigma S\sqrt{h}Z$. Since the random Brownian term is of order $\sqrt{h}$, the whole stock increment is dominated by $\delta S \sim O(\sqrt{h})$. Observe from our formulation of the full second-order Taylor expansion that we are comparing the terms h^2, $h\delta S$ and $(\delta S)^2$. First the P_{tt} term is propotional to h^2 so dividing by h we get that $h^2/h = h$ meaning that the limit $\lim_{h\to 0} = 0$. So, $h^2 = o(h)$ which is why $\frac{1}{2}P_{tt}h^2$ vanish relative to our time increment.

Consider the mixed Taylor term that contains $\delta t\delta S$ since $\delta S = \mu Sh + \sigma S\sqrt{h}Z$. Notice that expanding $h\delta S = \mu Sh^2 + \sigma Sh^{3/2}Z$ and dividing this by h gives us $\mu Sh + \sigma Sh^{1/2}Z$. Again, taking the limit $h \to 0$ we see that the two components vanish to 0. So, naturally the limit $\lim_{h\to 0}\frac{h\delta S}{h} = 0$ meaning that $h\delta S = o(h)$. Equivalently, since $\delta S \sim O(\sqrt{h})$ then $h\delta S \sim h\sqrt{h} = h^{3/2}$. Dividing this by h we see that this limit vanishes to 0 as well as $h \to 0$ so the mixed Hessian term $P_{tS}h\delta S$ also vanishes at the dt scale.

The most important term is the $(\delta S)^2$ term which if we compute directly we get,

$$\begin{aligned}(\delta S)^2 &= \left(\mu S h + \sigma S\sqrt{h}Z\right)^2 \\ &= \mu^2 S^2 h^2 + 2\mu\sigma S^2 h^{3/2} Z + \sigma^2 S^2 h Z^2.\end{aligned}$$

Dividing by h we get,

$$\begin{aligned}\frac{(\delta S)^2}{h} &= \frac{\mu^2 S^2 h^2 + 2\mu\sigma S^2 h^{3/2} Z + \sigma^2 S^2 h Z^2}{h} \\ &= \mu^2 S^2 h + 2\mu\sigma S^2 h^{1/2} Z + \sigma^2 S^2 Z^2.\end{aligned}$$

Taking the limit of the final equation above as $h \to 0$ we see that the first two terms vanish as well. However, notice that the third term does not vanish, that is $\sigma^2 S^2 Z^2$ survives the δt. This surviving contribution is summarized by the quadratic variation rule that $(\delta W^2) \sim h$ so we see that,

$$\begin{aligned}(\delta S)^2 &= (\mu S h + \sigma S \delta W)^2 \\ &= \mu^2 S^2 h^2 + 2\mu\sigma S^2 h \delta W + \sigma^2 S^2 (\delta W)^2 \\ &\approxeq 0 + 0 + \sigma^2 S^2 h \\ &= \sigma^2 S^2 h.\end{aligned}$$

So the continuous-time scaling is $h^2 = o(h)$ which means that $h\delta S = o(h)$ and $(\delta S)^2 = O(h)$.

Approximating Option Vega

The Black-Scholes vega-gamma identity is,

$$\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} = S^2 \hat{\sigma} T \frac{\partial^2 P_{\hat{\sigma}}}{\partial S^2}.$$

Recall our definitions for d_1 and d_2. For a call, Black-Scholes says that the option premium is,

$$P_{\hat{\sigma}} = S e^{-qT} N(d_1) - K e^{-rT} N(d_2).$$

Taking the first derivative wrt $\hat{\sigma}$ we get,

$$\begin{aligned}\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} &= S e^{-qT} \phi(d_1) \frac{\partial d_1}{\partial \hat{\sigma}} - K e^{-rT} \phi(d_2) \frac{\partial d_2}{\partial \hat{\sigma}} \\ &= S e^{-qT} \phi(d_1) \sqrt{T}.\end{aligned}$$

The gamma is therefore written as,

$$\frac{\partial^2 P_{\hat{\sigma}}}{\partial S^2} = \frac{\partial}{\partial S}\left(e^{-qT} N(d_1)\right) = e^{-qT}\phi(d_1)\frac{\partial d_1}{\partial S} = e^{-qT}\phi(d_1)\frac{1}{S\hat{\sigma}\sqrt{T}}.$$

Now we multiply the gamma be $S^2\hat{\sigma}T$,

$$\begin{aligned}S^2\hat{\sigma}T\frac{\partial^2 P_{\hat{\sigma}}}{\partial S^2} &= S^2\hat{\sigma}T\left(e^{-qT}\phi(d_1)\frac{1}{S\hat{\sigma}\sqrt{T}}\right) \\ &= S e^{-qT}\phi(d_1)\frac{\hat{\sigma}T}{\hat{\sigma}\sqrt{T}} = S e^{-qT}\phi(d_1)\sqrt{T} = \frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}}.\end{aligned}$$

Therefore we see that $\frac{\partial P_{\hat{\sigma}}}{\partial \hat{\sigma}} = S^2\hat{\sigma}T\frac{\partial^2 P_{\hat{\sigma}}}{\partial S^2}$ because the vega is $S e^{-qT}\phi(d_1)\sqrt{T}$ while the gamma is $\frac{e^{-qT}\phi(d_1)}{S\hat{\sigma}\sqrt{T}}$.

So multiplying the gamma by $S^2\hat{\sigma}T$ exactly converts it to vega.

Forward Approximations of Black-Scholes and Black-76

Consider the discount factor $D = e^{-rT}$ and log-moneyness of a forward contract at the forward price F and strike K given by $x = \log(F/K)$. For simplicity, we denote $a = \sigma\sqrt{T}$.

Under Black-Scholes/Black-76 forward dynamics we assume that,

$$F_T = F \exp\left\{-\frac{1}{2}\sigma^2\sqrt{T} + \sigma\sqrt{T}Z\right\}, \quad Z \sim \mathcal{N}(0,1).$$

Under the risk-neutral measure, the call and put prices are simply the discounted risk-neutral expectations,

$$C = D \cdot \mathbb{E}[(F_T - K)^+], \qquad P = D \cdot \mathbb{E}[(K - F_T)^+].$$

We refer to a call being "in-the-money" when,

$$\begin{aligned} F_T > K &\iff F \exp\left(-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}Z\right) > K \\ &\iff \log(F/K) - \frac{1}{2}\sigma^2 T + \sigma\sqrt{T}Z > 0 \\ &\iff Z > \frac{-\log(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}. \end{aligned}$$

We therefore define d_1 and d_2 as,

$$d_1 = \frac{\log(F/K) + \frac{1}{2}\sigma^2}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} = \frac{\log(F/K) - \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}.$$

The exercise condition is therefore when $F_T > K \iff Z > -d_2$.

We re-write the call formula as,

$$\begin{aligned} C &= D\mathbb{E}[(F_T - K)^+] \\ &= D\mathbb{E}[(F_T - K)\mathbf{1}\{F_T > K\}] \\ &= D\left(\mathbb{E}[F_T\mathbf{1}\{F_T > K\}] - K\mathbb{P}(F_T > K)\right). \end{aligned}$$

Since $F_T > K \iff Z > -d_2$ then $\mathbb{P}(F_T > K) = \mathbb{P}(Z > -d_2)$ which is just the normal CDF $\phi(d_2)$. Also, we substitute the dynamics of F_T into the expectation to get,

$$\mathbb{E}[F_T\mathbf{1}\{F_T > K\}] = F\mathbb{E}\left[e^{-\frac{1}{2}a^2 + aZ}\mathbf{1}\{Z > -d_2\}\right].$$

For a Gaussian, we can use the completing-square identity given as,

$$\begin{aligned} e^{-\frac{1}{2}a^2 + az}\phi(z) &= e^{-\frac{1}{2}a^2 + az}\frac{1}{\sqrt{2\pi}}e^{-z^2/2} \\ &= \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z^2 - 2az + a^2)} \\ &= \frac{1}{\sqrt{2\pi}}e^{-(z-a)^2/2} \\ &= \phi(z - a). \end{aligned}$$

By definition of expectations we simply compute,

$$\begin{aligned} \mathbb{E}\left[e^{-\frac{1}{2}a^2 + aZ}\mathbf{1}\{Z > -d_2\}\right] &= \int_{-d_2}^{\infty} e^{-\frac{1}{2}a^2 + az}\phi(z)dz \\ &= \int_{-d_2}^{\infty} \phi(z - a)dz \\ &= \int_{-d_2 - a}^{\infty} \phi(u)du \\ &= \phi(d_2 + a) = \phi(d_1). \end{aligned}$$

Therefore the Black-Scholes call price can be written as,

$$C_{\text{BC}} = D(F\phi(d_2) - K\phi(d_2)) = e^{-rT}(F\phi(d_1) - K\phi(d_2)).$$

When rT is small, the discount factor is close to 1, so the Black-Scholes call price is simply $F\phi(d_1) - K\phi(d_2)$. By put-call parity using our forward notation,

$$C_{\text{BS}} - P_{\text{BS}} = D(F - K) \implies P_{\text{BS}} = C_{\text{BS}} - D(F - K).$$

Substituting the formulation for the Black-Scholes call price,

$$\begin{aligned} P_{\text{BS}} &= D(F\phi(d_1) - K\phi(d_2)) - D(F - K) \\ &= D(F\phi(d_1) - K\phi(d_2) - F + K) \\ &= D(K(1 - \phi(d_2)) - F(1 - \phi(d_1))). \end{aligned}$$

By symmetry of the normal CDF density,

$$1 - \phi(x) = \phi(-x).$$

Therefore the Black-Scholes put price by put-call parity is simply,

$$P_{\text{BS}} = D(K\phi(-d_2) - F\phi(-d_1)) = e^{-rT}(K\phi(-d_2) - F\phi(-d_2)).$$

Some useful properties for deriving the greeks of options. Clearly $d_1 - d_2 = \sigma\sqrt{T} = a$ and,

$$d_1 + d_2 = \frac{2\log(F/K)}{\sigma\sqrt{T}} = \frac{2x}{a}.$$

The difference between the squares of the terms is simply,

$$\begin{aligned} d_1^2 - d_2^2 &= (d_1 - d_2)(d_1 + d_2) \\ &= a\frac{2x}{a} = 2x \\ &= 2\log(F/K). \end{aligned}$$

The ratio of the densities of the two terms is therefore simply,

$$\begin{aligned} \frac{\phi(d_1)}{\phi(d_2)} &= \frac{e^{-d_1^2/2}}{e^{-d_2^2/2}} \\ &= e^{-(d_1^2 - d_2^2)/2} \\ &= e^{-\log(F/K)} = \frac{K}{F}. \end{aligned}$$

Thus it is clear that $F\phi(d_1) = K\phi(d_2)$. Note the partial derivatives of our terms wrt the forward price are equal,

$$\frac{\partial d_1}{\partial F} = \frac{\partial d_2}{\partial F} = \frac{1}{F\sigma\sqrt{T}}.$$

Therefore, the derivatives of the call prices wrt F are,

$$\begin{aligned} \frac{\partial C}{\partial F} &= D\left[\phi(d_1) + F\phi(d_1)\frac{\partial d_1}{\partial F} - K\phi(d_2)\frac{\partial d_2}{\partial F}\right] \\ &= D\left[\phi(d_1) + (F\phi(d_1) - K\phi(d_2))\frac{1}{F\sigma\sqrt{T}}\right] \\ &= D\phi(d_1) = e^{-rT}\phi(d_1). \end{aligned}$$

By put-call parity the delta of a put is therefore $-D\phi(-d_1) = D(\phi(d_1) - 1)$.

Taking the derivative of the delta wrt to F we can get the Gamma of our option. Since the delta of a call option is simply $D\phi(d_1)$, the derivative of this identity measures the sensitivity of delta to moves in the underlying spot. It follows as,

$$\begin{aligned}\Gamma_F &= \frac{\partial^2 C}{\partial F^2} \\ &= \frac{\partial}{\partial F}(D\phi(d_1)) \\ &= D\phi(d_1)\frac{\partial d_1}{\partial F} \\ &= D\phi(d_1)\frac{1}{F\sigma\sqrt{T}} = e^{-rT}\frac{\phi(d_1)}{F\sigma\sqrt{T}}.\end{aligned}$$

Note that the derivative is the same as it is for puts as it is for calls,

$$\Gamma_F^C = \Gamma_F^P = \frac{e^{-rT}\phi(d_1)}{F\sigma\sqrt{T}}.$$

For short maturities, $rT \approx 0$ so $\Gamma_F \approx \frac{\phi(d_1)}{F\sigma\sqrt{T}}$.

Differentiating call prices wrt σ we get,

Autocorrelation term of PnL Variance

In Section 1.3, we introduce the normalized off-diagonal covariance of instantaneous variance where the off-diagonal term is $\mathbb{E}[\sigma_i^2\sigma_j^2] - \hat{\sigma}^4$. Suppose that we let $X_i := \sigma_i^2 z_i^2 - \hat{\sigma}^2$ and let $X_j := \sigma_j^2 z_j^2 - \hat{\sigma}^2$.

Their cross-moment can therefore be written as,

$$\begin{aligned}
\mathbb{E}[X_i X_j] &= \mathbb{E}[(\sigma_i^2 z_i^2 - \hat{\sigma}^2)(\sigma_j^2 z_j^2 - \hat{\sigma}^2)] \\
&= \mathbb{E}[\sigma_i^2 z_i^2 \sigma_j^2 z_j^2 - \hat{\sigma}^2 \sigma_i^2 z_i^2 - \hat{\sigma}^2 \sigma_j^2 z_j^2 - \hat{\sigma}^4] \\
&= \mathbb{E}[\sigma_i^2 z_i^2 \sigma_j^2 z_j^2] - \hat{\sigma}^2 \mathbb{E}[\sigma_i^2 z_i^2] - \hat{\sigma}^2 \mathbb{E}[\sigma_j^2 z_j^2] + \hat{\sigma}^4 \\
&= \mathbb{E}[\sigma_i^2 \sigma_j^2]\mathbb{E}[z_i^2 z_j^2] - \hat{\sigma}^2 \mathbb{E}[\sigma_i^2]\mathbb{E}[z_i^2] - \hat{\sigma}^2 \mathbb{E}[\sigma_j^2]\mathbb{E}[z_j^2] + \hat{\sigma}^4 \\
&= \mathbb{E}[\sigma_i^2 \sigma_j^2] \cdot (1)(1) - \hat{\sigma}^2 \mathbb{E}[\sigma_i^2] \cdot (1) - \hat{\sigma}^2 \mathbb{E}[\sigma_j^2] \cdot (1) + \hat{\sigma}^4 \\
&= \mathbb{E}[\sigma_i^2 \sigma_j^2] - \hat{\sigma}^4 - \hat{\sigma}^4 + \hat{\sigma}^4 \\
&= \mathbb{E}[\sigma_i^2 \sigma_j^2] - \hat{\sigma}^4 \\
&= \mathbb{E}[\sigma_i^2 \sigma_j^2] - \left(\hat{\sigma}^2\right)^2 \\
&= \mathrm{Cov}(\sigma_i^2, \sigma_j^2).
\end{aligned}$$

Normalizing the covariance by the variance of σ^2 we get,

$$f_{ij} = \frac{\mathrm{Cov}(\sigma_i^2, \sigma_j^2)}{\mathrm{Var}(\sigma^2)} = \frac{\mathbb{E}[\sigma_i^2 \sigma_j^2] - \hat{\sigma}^4}{\mathbb{E}[\sigma^4] - \sigma^4}.$$

$\mathbb{E}[\sigma_i^2\sigma_j^2]$ measures the expectation when the variance is high today, how much more likely is variance to be high tomorrow (or in the next time unit). Suppose first that volatility were completely deterministic i.e., at $\sigma_t = 20\%$ for every day. Then we would assume that $\sigma_i^2 = \sigma_j^2 = \hat{\sigma}^2$ always so $\mathbb{E}[\sigma_i^2\sigma_j^2] = \hat{\sigma}^4$ meaning that $\mathbb{E}[\sigma_i^2\sigma_j^2] - \hat{\sigma}^4 = 0$ i.e., there is absolutely no randomness in volatility itself. Consequently this means that our autocovariance term defined above would simply be zero. Imagine, on the contrary that the market behaves with high volatility days tending to occur together and low-volatility days likewise meaning that volatility has "memory". In the case that σ_i^2 is above average, then σ_j is expected to be above average as well provided that j is not too far away. Consequently, $\mathbb{E}[\sigma_i^2\sigma_j^2] > \mathbb{E}[\sigma_i^2]\mathbb{E}[\sigma_j^2] = \hat{\sigma}^4$ meaning that $\mathbb{E}[\sigma_i^2\sigma_j^2] - \hat{\sigma}^4 > 0$ representing the quantity of the covariance of volatility. Specifically, $\mathrm{Cov}(\sigma_i^2, \sigma_j^2)$ has units of variance squared so it depends on whether volatility is quoted in percentages or decimals making it difficult to compare different markets. Normalizing it solves this problem exactly as one does when defining an ordinary correlation coefficient. When $f_{ij} = 1$ then $\sigma_i^2 = \sigma_j^2$ perfectly meaning that knowing today's variance today tells you tomorrow's variance exactly. On the contrary if $f_{ij} = 0$ then $\sigma_i \neq \sigma_j$ meaning that there today's volatility tells you nothing about tomorrow's or the next units volatility.

Kolmogorov Equations

Forward Kolmogorov Equation

$$\begin{aligned}
\frac{\partial}{\partial t} p(s, x; t, y) &= \lim_{h \downarrow 0} \frac{p(s, x; t+h, y) - p(s, x; t, y)}{h} \\
&= \lim_{h \downarrow 0} \frac{1}{h} \left\{ \int_{\mathbb{R}} p(s, x; t, z) p(t, z; t+h, y)\, dz - p(s, x; t, y) \right\} \\
&= \lim_{h \downarrow 0} \frac{1}{h} \left\{ \int_{\mathbb{R}} p(s, x; t, z) \left[p(t, z; t+h, y) - \delta_y(z) \right]\, dz \right\} \\
&= \int_{\mathbb{R}} p(s, x; t, z) \lim_{h \downarrow 0} \frac{p(t, z; t+h, y) - \delta_y(z)}{h}\, dz \\
&= \int_{\mathbb{R}} p(s, x; t, z)\, \mathcal{L}_t^{*,y} \delta_y(z)\, dz \\
&= \mathcal{L}_t^{*,y} p(s, x; t, y).
\end{aligned}$$

The *Forward Kolmogorov Equation* is therefore,

$$\boxed{\frac{\partial}{\partial t}p(s,x;t,y) = \mathcal{L}_t^{*,y}p(s,x;t,y)}$$

Backward Kolmogorov Equation

$$\begin{aligned}
\frac{\partial}{\partial s}p(s,x;t,y) &= \lim_{h\downarrow 0}\frac{p(s+h,x;t,y)-p(s,x;t,y)}{h}\\
&= \lim_{h\downarrow 0}\frac{1}{h}\left\{p(s+h,x;t,y)-p(s,x;t,y)\right\}\\
&= \lim_{h\downarrow 0}\frac{1}{h}\left\{p(s+h,x;t,y)-\int_{\mathbb{R}}p(s,x;s+h,z)p(s+h,z;t,y)\,dz\right\}\\
&= \lim_{h\downarrow 0}\frac{1}{h}\left\{\int_{\mathbb{R}}\delta_x(z)p(s+h,z;t,y)\,dz-\int_{\mathbb{R}}p(s,x;s+h,z)p(s+h,z;t,y)\,dz\right\}\\
&= \lim_{h\downarrow 0}\int_{\mathbb{R}}\frac{\delta_x(z)-p(s,x;s+h,z)}{h}p(s+h,z;t,y)\,dz\\
&= -\int_{\mathbb{R}}\lim_{h\downarrow 0}\frac{p(s,x;s+h,z)-\delta_x(z)}{h}p(s,z;t,y)\,dz\\
&= -\int_{\mathbb{R}}\mathcal{L}_s^{*,z}\delta_x(z)p(s,z;t,y)\,dz\\
&= -\int_{\mathbb{R}}\delta_x(z)\mathcal{L}_s^{z}p(s,z;t,y)\,dz\\
&= -\mathcal{L}_s^{x}p(s,x;t,y)\\
&= -\left[b(s,x)\frac{\partial}{\partial x}p(s,x;t,y)+\frac{1}{2}\sigma^2(s,x)\frac{\partial^2}{\partial x^2}p(s,x;t,y)\right]\\
&= -b(s,x)\frac{\partial}{\partial x}p(s,x;t,y)-\frac{1}{2}\sigma^2(s,x)\frac{\partial^2}{\partial x^2}p(s,x;t,y).
\end{aligned}$$

The *Backward Kolmogorov Equation* is therefore,

$$\boxed{\frac{\partial}{\partial s}p(s,x;t,y) = -b(s,x)\frac{\partial}{\partial x}p(s,x;t,y)-\frac{1}{2}\sigma^2(s,x)\frac{\partial^2}{\partial x^2}p(s,x;t,y)}$$

Chapman-Kolmogorov

The Chapman-Kolmogorov equation says that if a Markov process starts at x at time s and ends at y at time t, and we force it to pass through some intermediate time u, then the total transition density from x to y is obtained by summing over every possible intermediate state z,

$$p(s,x;t,y) = \int_{\mathbb{R}}p(s,x;u,z)p(u,z;t,y)dz, \quad s<u<t.$$

That is, the probability of x going to y is the sum over the probabilities of x going to intermediate time(s) z and probability of z going to final time y. For a continuous state variable, the sum over z becomes an integral,

$$\begin{aligned}
p(s,x;t,y) &= \mathbb{P}(X_t\in[y,y+dy]\mid X=x)\\
&= \mathbb{P}(X_t\in[y,y+dy], X_u\in dz\mid X_s=z)\\
&= \int_{\mathbb{R}}\mathbb{P}(X_t\in[y,y+dy]\mid X_u=z, X_s=x)\mathbb{P}(X_u\in dz\mid X_s=x).
\end{aligned}$$

Applying the Markov property says that once we know that $X_u=z$, then the earlier value $X_s=x$ no longer matters for predicting X_t,

$$\mathbb{P}(X_t\in[y,y+dy]\mid X_u=z, X_s=x) = \mathbb{P}(X_t\in[y,y+dy]\mid X_u=z).$$

Therefore the transition state density becomes,

$$\begin{aligned} p(s,x;t,y)dy &= \int_{\mathbb{R}} \mathbb{P}(X_t \in [y, y+dy] \mid X_u = z)\mathbb{P}(X_u \in dz \mid X_s = z) \\ &= \int_{\mathbb{R}} p(u,z;t,y)dy \cdot p(s,x;u,z)dz \\ &= \left(\int_{\mathbb{R}} p(s,x;u,z)p(u,z;t,y)dz\right) dy. \end{aligned}$$

Dividing by dy we simply get,

$$p(s,x;t,y) = \int_{\mathbb{R}} p(s,x;u,z)p(u,z;t,y)dz.$$

So it simply follows for us to write,

$$\begin{aligned} \mathbb{P}(X_t \in dy \mid X_s = x) &= \int_{\mathbb{R}} \mathbb{P}(X_t \in dy, X_u \in dz \mid X_s = x) \\ &= \int_{\mathbb{R}} \mathbb{P}(X_t \in dy \mid X_u = z, X_s = x)\mathbb{P}(X_u \in dz \mid X_s = z) \\ &= \int_{\mathbb{R}} \mathbb{P}(X_t \in dy \mid X_u = z)\mathbb{P}(X_u \in dz \mid X_s = z) \\ &= \int_{\mathbb{R}} p(u,z;t,y)dy \cdot p(s,x;u,z)dz \\ &= \left(\int_{\mathbb{R}} p(s,x;u,z)p(u,z;t,y)dz\right) dy. \end{aligned}$$

Therefore , to get from x at time s to y at time t, the process must be somewhere z at the intermediate time u. The density of going from x to z is $p(s,x;u,z)$ so the density of then going from z to y is $p(u,z;t,y)$ and integrating over all possible z gives the full density.

2.6 Local-Volatility Price Dynamics

Suppose we let $\tilde{p}(0,T,x,y)$ represent the density of $S_T = y$ given that $S_0 = x$. Under local volatility we have that $dS_t = (r-q)S_t dT + \sigma_{\text{loc}}(T, S_T)S_T dW_T$. So if the terminal state variable is y, then the drift and variance coefficients are $\mu(T,y) = (r-q)y$ and $a(T,y) = \sigma^2_{\text{loc}}(T,y)y^2$. The forward kolmogorov equation says that the terminal density satisfies,

$$\tilde{p}_T = -\frac{\partial}{\partial y}(\mu\tilde{p}) + \frac{1}{2}\frac{\partial^2}{\partial y^2}(a\tilde{p}).$$

Suppose we have a tiny time step for small h,

$$\Delta S = S_{T+h} - S_T \approx \mu(T,y)h + \sqrt{a(T,y)}\sqrt{h}Z, \quad Z \sim \mathcal{N}(0,1).$$

For any smooth test function f we have,

$$\begin{aligned} \mathbb{E}[f(S_{T+h}) \mid S_T = y] &= \mathbb{E}[f(y + \Delta S)] \\ &= \mathbb{E}\left[f(y) + f'(y)\Delta S + \frac{1}{2}f''(y)(\Delta S)^2 + o(h)\right] \\ &= f(y) + f'(y)\mathbb{E}[\Delta S] + \frac{1}{2}f''(y)\mathbb{E}[(\Delta S)^2] + o(h). \end{aligned}$$

We compute the two moment as,

$$\begin{aligned} \mathbb{E}[\Delta S] &= \mathbb{E}[\mu h + \sqrt{a}\sqrt{h}Z] \\ &= \mu h + \sqrt{a}\sqrt{h}\mathbb{E}[Z] \\ &= \mu h, \end{aligned}$$

$$\begin{aligned}\mathbb{E}[(\Delta S)^2] &= \mathbb{E}\left[(\mu h + \sqrt{a}\sqrt{h}Z)^2\right] \\ &= \mathbb{E}\left[\mu^2 h^2 + 2\mu\sqrt{a}h^{3/2}Z + ahZ^2\right] \\ &= \mu^2 h^2 + 2\mu\sqrt{a}h^{3/2}\mathbb{E}[Z] + ah\mathbb{E}[Z^2] \\ &= \mu^2 h^2 + ah \\ &= ah + o(h).\end{aligned}$$

Therefore it follows that our smooth test function f has conditional expectation given by,

$$\mathbb{E}[f(S_{T+h}) \mid S_T = y] = f(y) + \mu h f'(y) + \frac{1}{2}ahf''(y) + o(h).$$

Subtracting $f(y)$ and dividing by h letting $h \to 0$ we see that asymptotically,

$$\lim_{h\to 0} \frac{\mathbb{E}[f(S_{T+h}) \mid S_T = y] - f(y)}{h} = \mu(T,y)f'(y) + \frac{1}{2}a(T,y)f''(y).$$

Using the densities we can obtain the expectation at time T which is,

$$\mathbb{E}[f(S_T)] = \int_0^\infty f(y)\tilde{p}(0,T,x,y)dy.$$

Differentiating this wrt T we get that,

$$\begin{aligned}\frac{\partial}{\partial T}\mathbb{E}[f(S_T)] &= \frac{\partial}{\partial T}\int_0^\infty f(y)\tilde{p}(0,T,x,y)dy \\ &= \int_0^\infty f(y)\tilde{p}_T(0,T,x,y)dy.\end{aligned}$$

However, observe that from the generator of the forward Kolmogorov equation we have that,

$$\begin{aligned}\frac{\partial}{\partial T}\mathbb{E}[f(S_T)] &= \mathbb{E}[\mathcal{L}f(S_T)] \\ &= \int_0^\infty \left[\mu(T,y)f_y(y) + \frac{1}{2}a(T,y)f_{yy}(y)\right]\tilde{p}(0,T,x,y)dy.\end{aligned}$$

It follows that we have that,

$$\int_0^\infty f(y)\tilde{p}_T dy = \int_0^\infty \mu f_y \tilde{p} dy + \frac{1}{2}\int_0^\infty a f_{yy}\tilde{p}dy.$$

We integrate by parts for the drift, diffusion (twice),

$$\begin{aligned}\int_0^\infty \mu f_y \tilde{p}\, dy &= \int_0^\infty f_y(\mu\tilde{p})\, dy \\ &= [f(\mu\tilde{p})]_0^\infty - \int_0^\infty f\frac{\partial}{\partial y}(\mu\tilde{p})\, dy = -\int_0^\infty f\frac{\partial}{\partial y}(\mu\tilde{p})\, dy.\end{aligned}$$

$$\begin{aligned}\frac{1}{2}\int_0^\infty a f_{yy}\tilde{p}\, dy &= \frac{1}{2}\int_0^\infty f_{yy}(a\tilde{p})\, dy \\ &= \frac{1}{2}\left[f_y(a\tilde{p})\right]_0^\infty - \frac{1}{2}\int_0^\infty f_y\frac{\partial}{\partial y}(a\tilde{p})\, dy \\ &= -\frac{1}{2}\int_0^\infty f_y\frac{\partial}{\partial y}(a\tilde{p})\, dy \\ &= -\frac{1}{2}\left[f\frac{\partial}{\partial y}(a\tilde{p})\right]_0^\infty + \frac{1}{2}\int_0^\infty f\frac{\partial^2}{\partial y^2}(a\tilde{p})\, dy = \frac{1}{2}\int_0^\infty f\frac{\partial^2}{\partial y^2}(a\tilde{p})\, dy.\end{aligned}$$

Therefore it follows that we have,

$$\begin{aligned}\int_0^\infty f(y)\tilde{p}_T\,dy &= -\int_0^\infty f(y)\frac{\partial}{\partial y}(\mu\tilde{p})\,dy + \frac{1}{2}\int_0^\infty f(y)\frac{\partial^2}{\partial y^2}(a\tilde{p})\,dy \\ &= \int_0^\infty f(y)\left[-\frac{\partial}{\partial y}(\mu\tilde{p}) + \frac{1}{2}\frac{\partial^2}{\partial y^2}(a\tilde{p})\right]dy.\end{aligned}$$

Since this holds for every smooth test function f, the integrands must be equal meaning that,

$$\tilde{p}_T = -\frac{\partial}{\partial y}(\mu\tilde{p}) + \frac{1}{2}\frac{\partial^2}{\partial y^2}(a\tilde{p}).$$

Finally substituting $\mu(T,y) = (r-q)y$ and $a(T,y) = \sigma^2_{\text{loc}}(T,y)y^2$ we get,

$$\tilde{p}_T(0,T,x,y) = -\frac{\partial}{\partial y}\left((r-q)y\,\tilde{p}(0,T,x,y)\right) + \frac{1}{2}\frac{\partial^2}{\partial y^2}\left(\sigma^2_{\text{loc}}(T,y)\,y^2\,\tilde{p}(0,T,x,y)\right).$$

This is precisely the forward kolmogorov equation for the density of S_T.

Second-Order Diffusion Reduction Identity

Suppose we let $A(y) := \sigma^2_{\text{loc}}(T,y)y^2\tilde{p}(0,T,x,y)$ so that component (2) in section 2.1 becomes,

$$\frac{1}{2}\int_K^\infty (y-K)A''(y)dy.$$

Now we apply integration by parts once,

$$\begin{aligned}\frac{1}{2}\int_K^\infty (y-K)A''(y)dy &= \frac{1}{2}[(y-K)A'(y)]_K^\infty - \frac{1}{2}\int_K^\infty A'(y)\frac{\partial}{\partial y}(y-K)dy \\ &= \frac{1}{2}[(y-K)A'(y)]_K^\infty - \frac{1}{2}\int_K^\infty A'(y)dy \\ &= \left\{\lim_{y\to\infty}(y-K)A'(y) - (K-K)A'(K)\right\} - \frac{1}{2}\int_K^\infty A'(y)dy \\ &= 0 - \frac{1}{2}\int_K^\infty A'(y)dy.\end{aligned}$$

Now it remains to integrate the derivative which simply becomes,

$$\begin{aligned}-\frac{1}{2}\int_K^\infty A'(y)dy &= -\frac{1}{2}[A(y)]_K^\infty \\ &= -\frac{1}{2}\left(\lim_{y\to\infty}A(y) - A(K)\right) \\ &= -\frac{1}{2}(0 - A(K)) = \frac{1}{2}A(K).\end{aligned}$$

Substituting $A(K)$ back into the formulation we get that,

$$\frac{1}{2}A(K) = \frac{1}{2}\sigma^2_{\text{loc}}(T,K)K^2\tilde{p}(0,T,x,K).$$